



HYBRID MODELING OF LONG-TERM ELECTRICITY DEMAND IN NEW SOUTH WALES USING PROPHET, SARIMA, AND XGBOOST

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August 2025

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS OF THE DEGREE OF
MASTER OF STATISTICS

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Acknowledgements

At the completion of this thesis, I sincerely thank all those who have supported and helped me during the research process.

First of all, I would like to express my special gratitude to my supervisor, Professor Wei Tian, for his meticulous guidance and selfless assistance throughout my entire research process. From topic selection, theoretical framework construction to empirical analysis, every step was inseparable from the patient and meticulous guidance and encouragement of my supervisor, which benefited me a lot.

Secondly, I would like to express my gratitude to UNSW for providing a favorable research environment and resource support, which has offered a solid guarantee for the smooth progress of my research. At the same time, I would also like to express my gratitude to all the course instructors for the inspiration and knowledge accumulation they have provided during the course study, which have laid a solid foundation for the completion of this thesis.

I would like to express my gratitude to my fellow students and friends. Thank you for the help and encouragement you gave me when I encountered difficulties, which kept me motivated to move forward on my research path.

Finally, I would like to express my gratitude to my family. Thank you for always providing me with the strongest support and the deepest love. It is your understanding and support that have enabled me to study and focus on my research with peace of mind.

With this article, I would like to express my most sincere gratitude to all those who have helped and cared for me.

Junheng Tan, Muchen Wei, Sishi Zhang, Zhonghan He, Justin Jacob, 05/08/2025.

Abstract

Accurate forecasting of electricity demand plays a crucial role in energy planning, policy formulation, and infrastructure development. This study proposes a three-stage hybrid forecasting framework to predict long-term electricity demand in New South Wales, Australia. The framework decomposes the original time series into trend, seasonal, and residual components, which are separately modeled using Prophet, SARIMA, and XGBoost respectively. To validate the model's effectiveness, the proposed hybrid model is compared against several Individual models across different methodological categories, including traditional time series models (Prophet, SARIMA), machine learning (XGBoost), and neural networks (MLP, LSTM). Evaluation metrics such as RMSE and MAPE reveal that the hybrid model significantly outperforms the individual models, achieving the lowest RMSE (143,741.7) and MAPE (2.32%). The results highlight the advantages of combining complementary modeling techniques to capture different patterns in electricity demand data.

Contents

Chapter 1	Introduction	1
1.1	Limitations of Existing Methods	2
1.2	Hybrid Modeling Approaches	2
1.3	Research Objectives and Contributions	2
Chapter 2	Literature Review	4
2.1	Individual Models for Electricity Demand Forecasting	4
2.2	Trend Forecasting and Structural Variables	4
2.3	Seasonal Modeling with SARIMA	5
2.4	Residual Modeling Using Machine Learning	5
2.5	Benchmarking Hybrid Forecasting Models	5
2.6	Summary	6
Chapter 3	Overview of the Three-Stage Hybrid Framework	7
3.1	Overview of the Three-Stage Hybrid Framework	7
3.1.1	Stage One: Trend Modeling with Prophet	7
3.1.2	Stage Two: Seasonal Component Modeling with SARIMA	7
3.1.3	Stage Three: Residual Modeling with XGBoost	8
3.1.4	Summary	8
3.2	Stage 1: Long-term Trend Extraction using Prophet	8
3.2.1	Prophet Model Description	8
3.2.2	Strengths and Limitations	10
3.3	Stage 2: Seasonal Component Modeling using SARIMA	10
3.3.1	SARIMA Model Description	10
3.3.2	Strengths and Limitations	11
3.3.3	Autocorrelation Function	11
3.4	Stage 3: Residual Modeling using XGBoost	12
3.4.1	XGBoost Model Description	12
3.4.2	Interpretability with SHAP	13
3.5	Model Evaluation Metrics	13
Chapter 4	Data Description and Exploratory Data Analysis	15
4.1	Data Sources and Description	15
4.2	Preprocessing and Frequency Alignment	15
4.3	Variable Overview	17
4.4	Training and Testing Split	22
Chapter 5	Results and Discussion	24
5.1	Stage One: Trend Extraction Using Prophet	24

5.2	Stage Two: Seasonal Component Modeling with SARIMA	28
5.2.1	Residual Diagnostics of SARIMA Model via ACF	29
5.2.2	Interpretation of RMSE Comparison between SARIMA and Hybrid Forecasts	30
5.2.3	Applied Extracting Trend and Forecasting Seasonal	30
5.3	Stage Three: Residual Forecast using XGBoost	31
5.3.1	XGBoost Hyperparameter Tuning Strategy	31
5.3.2	Residual Modeling and Final Forecast Evaluation	35
5.4	Final Hybrid Forecast Evaluation	37
5.4.1	Model Interpretation with SHAP	38
5.4.2	Comparison Between Each Stage of Hybrid Models	40
5.4.3	Comparison Between Individual and Hybrid Models	41
5.5	Further Analysis	43
5.5.1	Performance of Different Season	43
5.5.2	Monthly APE Trend Analysis	45
Chapter 6	Conclusion and Future Work	47
6.1	Conclusion	47
6.2	Limitations and Future Work	47
	References	49

CHAPTER 1

Introduction

Due to the advancement of electrification, the implementation of climate policies, and the accelerated integration of renewable energy, the global energy landscape is undergoing a rapid transformation. Against this backdrop, accurate long-term **monthly** electricity demand forecasting has become the core support for energy planning, investment decisions, and supply-demand balance in a low-carbon future. As one of the major economic regions in Australia, New South Wales (NSW) presents unique challenges and opportunities in electricity forecasting. Factors such as the continuous growth of its population, the constant evolution of infrastructure, and the variable climate patterns all make the forecasting issue more complex.

In response to these challenges, this study seeks to answer the central research question: *How can we improve the accuracy and interpretability of long-term electricity demand forecasting in New South Wales by leveraging a task-decomposed hybrid modeling framework?*

Electricity demand time series typically exhibit multiple interacting temporal patterns, including long-term structural trends, cyclical seasonal variations, and short-term irregular fluctuations. Modeling all of these patterns jointly may obscure their individual dynamics and reduce forecast interpretability.

As emphasized by Hyndman and Athanasopoulos (2021) [1], many real-world time series can be effectively decomposed into three distinct components: a trend component that captures structural changes over time, a seasonal component that reflects recurring patterns, and a residual component that accounts for irregular variation. This decomposition not only enhances modeling clarity and precision but also allows each component to be modeled using techniques best suited to its statistical characteristics.

Therefore building on this foundation, this project adopts a decomposition-based approach that explicitly separates the time series into three parts: **trend**, **seasonal**, and **residual** components. Each component is modeled using a method that matches its statistical characteristics:

- **Trend Component:** Captures long-term socio-economic developments such as population growth, urbanization, and industrial activity. Extracting this component allows us to isolate the underlying direction of electricity consumption.
- **Seasonal Component:** Represents recurring and periodic fluctuations driven by temperature cycles. This part is well modeled by SARIMA, which handles autoregressive seasonal structures effectively.

- **Residual Component:** Contains short-term, irregular, or nonlinear effects that cannot be explained by trend or seasonality—often driven by high-dimensional external factors. XGBoost is suitable here for its ability to handle nonlinear interactions.

By structuring the task in this way, the framework leverages the strengths of different modeling techniques and enhances both predictive performance and interpretability. This approach also aligns with theoretical foundations of time series decomposition and addresses practical forecasting needs.

1.1 Limitations of Existing Methods

Traditional statistical models such as ARIMA introduced by Hyndman and Athanasopoulos (2021) [1] and SARIMA introduced by Box et al. (2015) [4] are widely used in time series forecasting due to their mathematical interpretability and ability to characterize time-dependent structures. However, when applied to long-term electricity demand forecasting, these models often struggle to cope with structural changes and nonlinear characteristics. In recent years, machine learning (ML) techniques, especially ensemble learning methods like XGBoost introduced by Chen and Guestrin (2016) [3], have demonstrated stronger predictive capabilities in modeling complex nonlinear relationships and high-dimensional interactions. However, machine learning models usually lack good interpretability, especially when it comes to decomposing trend and seasonal components.

1.2 Hybrid Modeling Approaches

To make up for the above deficiencies, structural decomposition methods like Prophet introduced by Taylor and Letham (2018) [2] have gradually become powerful modeling tools that can clearly break down trends, seasonality and holiday effects in time series. Prophet has good robustness against missing values and structural turning points, and is particularly suitable for long-term prediction tasks. However, due to the limitations of its additive structure, Prophet has difficulty modeling seasonal autoregressive patterns or complex residual structures. Therefore, in recent years, researchers have proposed an increasing number of hybrid modeling frameworks that combine statistical models with machine learning methods, but most methods either lack a clear task decomposition logic or are limited to long-term prediction tasks.

1.3 Research Objectives and Contributions

This project proposes a three-stage hybrid modeling framework, specifically designed for long-term (monthly) electricity demand forecasting in New South Wales. This method structurally decomposes the forecasting task into three stages: The first stage extracts long-term trends through Prophet; The second stage uses SARIMA to model seasonal fluctuations; In the third stage, multiple external variables are combined to model the residual structure using XGBoost. Furthermore the model introduces SHAP (Shapley Additive Explanation Method) in the final stage to enhance interpretability and reveal the importance of variables.

The contributions of this project are mainly reflected in three aspects:

- A modular and highly interpretable hybrid modeling framework is proposed, which is suitable for long-term electricity demand forecasting and can simultaneously capture structural changes and nonlinear features;
- This framework was applied to the real electricity demand data of NSW from 1999 to 2024, integrating exogenous variables such as economy, climate and technology.
- By leveraging SHAP analysis, the interpretability of the model output is achieved, enabling policymakers to clearly understand the impact of different variables on residual prediction.

CHAPTER 2

Literature Review

Electricity demand forecasting plays a crucial role in energy system planning, especially in the face of increasing electrification, climate change, and technological advancement. To enhance the accuracy and interpretability of predictions, recent research has increasingly adopted hybrid prediction models that decompose time series into trend, seasonal, and residual components. This project employs a three-stage hybrid modeling framework: Prophet is used to model long-term trends, SARIMA captures seasonal fluctuations, and XGBoost is employed to correct residuals. This approach is developed based on a large body of existing literature and aims to integrate the interpretability of statistical models with the predictive power of machine learning.

2.1 Individual Models for Electricity Demand Forecasting

Traditional statistical models such as ARIMA and SARIMA are widely used due to their ability to capture autocorrelation and periodic patterns in time series data. For example, Taylor and McSharry (2007) [5] applied SARIMA to model daily electricity demand in the UK, achieving reasonable accuracy but highlighting its limitations in handling structural changes. Similarly, the Prophet model developed by Facebook, has been applied extensively due to its robustness in managing trend shifts and missing data. However, it assumes an additive structure and may struggle to capture autocorrelated seasonality.

In the domain of machine learning, XGBoost has attracted widespread attention due to its strong predictive accuracy and robustness to outliers. As demonstrated by Chen and Guestrin (2016) [6], XGBoost performs exceptionally well in various regression tasks, including electricity demand forecasting. Nonetheless, its black-box nature limits interpretability, especially in contexts where decomposed trend and seasonal structures are essential.

2.2 Trend Forecasting and Structural Variables

For the long-term trend modeling part, it is crucial to identify structural changes in energy consumption. The Prophet model has been widely adopted because it can capture trend changes and handle missing data. Bedi and Toshniwal (2019) [7] proposed the DFED framework, which combines seasonal clustering with MIMO LSTM models to simultaneously capture slow changes and sudden fluctuations. Chen et al. (2018) [9] utilized SVR and random forest models to predict electricity demand in Guangdong, China, emphasizing the importance of regional macroeconomic variables like CPI.

Inspired by these approaches, our model introduces macroeconomic indicators (e.g., GSP) and external regional data (e.g., NSW data) into the trend modeling stage. This logic is also supported by the Australian Energy Market Operator (AEMO) (2023) [8], which incorporates structural variables into its long-term scenario forecasts.

2.3 Seasonal Modeling with SARIMA

SARIMA remains one of the most effective models for capturing periodic seasonal fluctuations and autocorrelation. Kumar Dubey et al. (2021) [10] compared the performance of ARIMA, SARIMA, and LSTM in household electricity demand and found SARIMA to be particularly robust when data volume was limited. Park and Yang (2024) [11] further emphasized SARIMA’s strength in modeling stable seasonal patterns.

In our model, SARIMA is simplified to focus solely on extracting seasonal structures after the trend component is removed, avoiding overfitting residuals. This separation allows each model to specialize in its domain.

2.4 Residual Modeling Using Machine Learning

Even after removing trend and seasonal components, electricity demand series may exhibit nonlinear residuals due to unpredictable factors like weather or behavioral shifts. XGBoost is employed in our model to handle these remaining nonlinearities. Mahesh et al. (2021) [12] proposed a hybrid ANN-LSTM model with PCA and feature selection, while Lu et al. (2020) [13] developed an LSTM-FCNN framework for correcting real-time fluctuations.

Our XGBoost-based residual model integrates contextual features like temperature and GSP to enhance performance. Compared to traditional tree-based models, XGBoost offers better accuracy and generalization ability in this context.

2.5 Benchmarking Hybrid Forecasting Models

Recent short-term electricity demand forecasting studies have provided valuable methodological inspiration for hybrid model design. Zeng et al. (2025) [14] introduced a Prophet-XGBoost hybrid model with Bayesian tuning, achieving a MAPE of 1.35% in Beijing. González Grandón et al. (2023) [15] proposed a macro-ARIMA-LSTM model for Ukraine’s grid, and Athanasiopoulou et al. (2022) [16] developed the HYENA model that integrates linear seasonality and LSTM autoencoders. Zhu et al. (2021) [17] also applied a Prophet-XGBoost hybrid model. While these models focus on short-term forecasting, their modular hybrid structures—typically dividing the task into trend, seasonal, and residual components—offer a useful reference for designing robust long-term models.

In contrast, long-term electricity demand forecasting faces unique challenges such as structural shifts, policy interventions, and macroeconomic fluctuations. Several studies have proposed hybrid frameworks to improve long-range accuracy. For example, Athanasiopoulou et al.’s (2022) [16] HYENA model achieved a MAPE of 3.33% in the UK. Eshragh et al. (2021) [27] used a SARIMA-regression hybrid with exogenous variables such as temperature and solar exposure, yielding a MAPE of 2.48%. Compared to these benchmarks, our three-stage Prophet-SARIMA-XGBoost

model achieves a lower MAPE of 2.32% for New South Wales, demonstrating superior long-term forecasting performance. Furthermore, unlike most prior studies, our model incorporates SHAP analysis to enhance interpretability.

We choose to use only the Mean Absolute Percentage Error (MAPE) to compare the performance of the models in the literature mentioned above. Compared with the Root Mean Square Error (RMSE), MAPE, as a relative error indicator, does not rely on the unit and dimension of the data, and thus is more suitable for horizontal comparison in different regions and under different data scales. Due to the fact that the prediction targets and regional scopes adopted in other literatures may vary, the RMSE values lack a unified scale basis and thus cannot serve as a reliable basis for comparison.

2.6 Summary

In summary, the existing literature supports the modular decomposition of time series for electricity forecasting. Although individual models offer simplicity and speed, they are often insufficient for complex and long-term electricity forecasting. Hybrid models address these limitations by combining the strengths of different modeling paradigms. Our study contributes by proposing a comprehensive three-stage hybrid model that effectively balances accuracy and interpretability for long-term monthly electricity demand forecasting in New South Wales.

CHAPTER 3

Overview of the Three-Stage Hybrid Framework

3.1 Overview of the Three-Stage Hybrid Framework

In order to improve the accuracy and interpretability of long-term electricity demand forecasting for New South Wales (NSW), this project proposes a task-level hybrid modeling framework, which decomposes the original time series into three independent components: trend, seasonality, and residuals. Each part adopts a specialized modeling method suitable for its characteristics, thereby overcoming the limitations of an individual model while combining the advantages of different algorithms.

To ensure the objectivity and generalizability of model evaluation, the time series data are divided into a training set and a testing set. Specifically, data from 1999 to 2022 are used to train the models, while data from 2023 to 2024 are reserved for testing forecasting capability. This partitioning is consistently applied across all three stages of the hybrid framework.

3.1.1 Stage One: Trend Modeling with Prophet

The Prophet model is used to fit the training set and extract the long-term trend components. The fitted model is then applied to the test set to predict trend changes. After obtaining the trend component, the RMSE and MAPE are calculated in both training and testing sets to evaluate whether overfitting has occurred. Furthermore, the Augmented Dickey-Fuller (ADF) test [18] is used to test the stationarity of the detrended series, ensuring that it is suitable for seasonal component modeling in the second stage.

3.1.2 Stage Two: Seasonal Component Modeling with SARIMA

The residuals from the first stage (i.e., the difference between the actual values y and the estimated trend $\hat{y}_{\text{trend}}$) are taken as input to model seasonal fluctuations using the SARIMA model. To constrain the model to seasonal patterns only, the parameters are stated and restricted as follows:

- d represents the non-seasonal differencing order, indicating how many times the data needs to be differenced overall to achieve stationarity. Therefore, $d = 0$ means no non-seasonal differencing is applied, assuming the trend has already been captured by Prophet;
- D represents the seasonal differencing order indicating whether differencing is needed at seasonal intervals, so $D = 1$ means seasonal differencing is applied to extract seasonal patterns;

- $max.p$ represents the maximum non-seasonal AR (AutoRegressive) order. It controls the maximum number of non-seasonal AR lags. Therefore, $max.p = 0$ means non-seasonal AR terms are not used;
- $max.q$ represents maximum non-seasonal MA (Moving Average) order. It controls the maximum number of non-seasonal MA lags. Therefore, $max.q = 0$ means non-seasonal MA terms are not used;
- $max.P$ represents the maximum seasonal AR order. It controls the number of seasonal AR lags allowed. Therefore, $max.P$ means at most one seasonal AR term is allowed;
- $max.Q$ represents the maximum seasonal MA order. It controls the number of seasonal MA lags allowed. Therefore, $max.Q = 1$ means at most one seasonal MA term is allowed.

So in this part, the parameters is set: $d = 0$, $D = 1$, $max.p = 0$, $max.q = 0$, $max.P = 1$, $max.Q = 1$.

The constrained SARIMA model is fitted on the training set and used to forecast the seasonal component in the test set. These forecasts are then combined with the trend forecasts to generate total electricity demand predictions. RMSE is computed to assess improvement from the first stage. The residuals from this stage are verified to be approximately white noise using the ACF residual plot, providing a theoretical basis for the third stage. This process yields the estimated seasonal component $\hat{y}_{\text{seasonal}}$.

3.1.3 Stage Three: Residual Modeling with XGBoost

The remaining residuals, defined as the difference between the actual values y and the combined outputs of trend and seasonal components ($y - \hat{y}_{\text{trend}} - \hat{y}_{\text{seasonal}}$), are modeled using the XGBoost algorithm. The dataset, which includes multiple exogenous variables, is standardized prior to modeling. XGBoost is trained on the training set, with 10-fold cross-validation used to enhance the model's generalization ability. The model is then applied to the test set to predict residual values, which are added to the corresponding trend and seasonality forecasts to form the final electricity demand prediction:

$$\hat{y} = \hat{y}_{\text{trend}} + \hat{y}_{\text{seasonal}} + \hat{y}_{\text{residual}}$$

Finally, the RMSE and MAPE of the final predicted electricity demand are calculated on both the training and testing sets to evaluate overall model performance.

3.1.4 Summary

This task-level hybrid structure ensures that each modeling task is optimally handled by a model tailored to its specific nature, thereby enhancing both forecasting accuracy and model interpretability.

3.2 Stage 1: Long-term Trend Extraction using Prophet

3.2.1 Prophet Model Description

The Prophet model, introduced by Taylor and Letham (2018) [2] from Facebook in 2018, is a versatile time series decomposition algorithm. Built upon an additive

structural modeling approach, it offers strong flexibility, interpretability, and automated capabilities. This model is particularly effective for datasets exhibiting clear trends, seasonality, holiday-related variations, and error terms.

In this project, the Prophet model is applied to analyze the trend patterns and structural shifts in long-term electricity demand within New South Wales (NSW). A key strength of the model is its capacity to detect underlying structural change points, adaptively fit trends in segmented intervals, and effectively manage missing data and outliers.

Specifically, Prophet adopts a generalized additive modeling approach, where any time series is expressed as the sum of the following components:

$$\hat{y}(t) = g(t) + s(t) + h(t) + \varepsilon_t \quad (3.2.1)$$

where:

- $\hat{y}(t)$ represents the predicted value at time t ;
- $g(t)$ is the non-periodic trend component;
- $s(t)$ is the seasonality component;
- $h(t)$ is the effect of holidays or special events;
- ε_t is the error term assumed to follow a normal distribution.

Trend Component $g(t)$

The trend term $g(t)$ models non-periodic long-term changes in the time series. It supports two trend forms:

- *Linear Growth Model (default):*

$$g(t) = k_{\text{adj}}t + m_{\text{adj}} \quad (3.2.2)$$

where k_{adj} is the adjusted growth rate and m_{adj} is the offset parameter at a changepoint. Prophet automatically detects multiple changepoints, allowing the model to apply piecewise linear trends over different time intervals.

- *Logistic Growth Model (for saturating trends):*

$$g(t) = \frac{C(t)}{1 + e^{-k_{\text{adj}}(t - m_{\text{adj}})}} \quad (3.2.3)$$

where $C(t)$ represents the capacity or upper limit.

Seasonality Component $s(t)$

The seasonality term $s(t)$ captures periodic fluctuations in the data caused by cyclical patterns such as weather and economic cycles. Prophet models this component using a Fourier series expansion:

$$s(t) = a_1 \cos\left(\frac{2\pi t}{P}\right) + b_1 \sin\left(\frac{2\pi t}{P}\right) + \dots + a_N \cos\left(\frac{2\pi Nt}{P}\right) + b_N \sin\left(\frac{2\pi Nt}{P}\right) \quad (3.2.4)$$

where P is the seasonality period (e.g., 12 for monthly data or 4 for quarterly data), and N is the number of Fourier terms. The coefficients a_N and b_N are estimated from the data.

Holiday Component $h(t)$

The holiday component $h(t)$ models the influence of specific known events such as public holidays, national events or energy policy announcements, which may cause temporary but significant deviations in electricity demand. Prophet models this component as:

$$h(t) = W(t)\kappa, \quad \kappa \sim \mathcal{N}(0, \sigma_\kappa^2) \quad (3.2.5)$$

where $W(t)$ is a binary vector indicating the occurrence of predefined holidays, and κ represents the strength of the holiday effect, drawn from a normal distribution prior.

3.2.2 Strengths and Limitations

Prophet offers excellent interpretability and robustness in handling time series with trend and seasonal structures. Key strengths include:

- Automatic detection of changepoints and flexible modeling of segmented trends;
- Handling of missing values and outliers with built-in robustness;
- Straightforward parameterization, suitable for business or engineering applications.

However, the model also has limitations:

- It assumes an additive structure, making it less suitable for data with multiplicative or strongly nonlinear interactions;
- Limited capability in handling exogenous variables or complex multivariate dependencies;
- Performance may deteriorate with sparse or irregularly distributed data;
- In scenarios with frequent structural breaks, changepoint detection may be suboptimal.

Therefore, while Prophet is an effective model for trend extraction, it should be complemented with additional modeling stages for components such as seasonality and residuals.

3.3 Stage 2: Seasonal Component Modeling using SARIMA

3.3.1 SARIMA Model Description

The Seasonal Autoregressive Integrated Moving Average (SARIMA) model is widely used for analyzing and forecasting time series with seasonal characteristics [19, 20, 21]. It is suitable not only for seasonal series but also for those with long-term trends. The SARIMA model is known for its accuracy and robustness in capturing seasonal structures. By differencing a non-stationary time series [22], it transforms it into a stationary time series that reveals trend and cyclical patterns, enabling reliable future forecasting.

Given a non-stationary series y_t with seasonality of period s , the d -order differencing can be expressed using the lag operator L as:

$$(1 - L)^d y_t = y_t - y_t^d \quad (3.3.1)$$

where L is the lag operator, and d is the number of non-seasonal differences.

The general ARIMA(p, d, q) model is expressed as:

$$(1 - \phi_1 L - \dots - \phi_p L^p)(1 - L)^d y_t = (1 + \theta_1 L + \dots + \theta_q L^q) \varepsilon_t \quad (3.3.2)$$

where:

- p is the autoregressive order; $\phi_1, \dots, \phi_p$ are AR coefficients;
- q is the moving average order; $\theta_1, \dots, \theta_q$ are MA coefficients;
- ε_t is a white noise error term.

To handle seasonal effects, the model is extended to SARIMA(p, d, q)(P, D, Q) $_s$, where s is the seasonal period. The full SARIMA model is written as:

$$\phi_p(L)A_p(L^s)(\Delta^d \Delta_s^D y_t) = \Theta_Q(L)B_Q(L^s)\varepsilon_t \quad (3.3.3)$$

where:

- $\phi_p(L)$ and $\theta_q(L)$ represent the non-seasonal and seasonal autoregressive polynomials, respectively;
- $A_P(L^s)$ and $B_Q(L^s)$ represent the non-seasonal and seasonal moving average polynomials;
- Δ^d and Δ_s^D denote the non-seasonal and seasonal difference operators;
- P, D , and Q represent the orders of non-seasonal and seasonal differencing, respectively, ensure that the original series $\{y_t\}$ is transformed into a stationary time series.

3.3.2 Strengths and Limitations

The SARIMA model offers several strengths:

- Effectively captures both seasonal and non-seasonal patterns in time series;
- Well-suited for datasets with regular periodicity and long historical records;
- Provides clear statistical interpretation and solid theoretical foundations.

However, it also has limitations:

- Requires the time series to be stationary, often necessitating careful differencing;
- Parameter selection (p, d, q, P, D, Q, s) can be complex and computationally intensive;
- Less effective when the data contain nonlinear structures or external regressors.

3.3.3 Autocorrelation Function

The autocorrelation function (ACF) is used to measure the correlation between a time series and its own different lag orders. In residual diagnosis, the ACF plot can help determine whether the residuals still have temporal dependence or structural patterns. For a well-fitted model, ideally, its residuals should exhibit white noise characteristics, that is, the autocorrelation coefficients of all lag orders should fall within the 95% confidence interval. If there are autocorrelation peaks that significantly exceed the confidence interval, it may indicate that the model has underfitting or there are still uncaptured time structures.

3.4 Stage 3: Residual Modeling using XGBoost

3.4.1 XGBoost Model Description

As an ensemble learning method, the **XGBoost** model optimizes the computing speed and performance of the Gradient Boosted Trees (GBT) algorithm. By using the gradient descent method to strengthen weak learners, it is an efficient implementation of the gradient boosting algorithm [23]. Compared with other gradient boosting algorithms, XGBoost can provide more accurate prediction nodes while accelerating computation.

For a given training set N , select a sample from (x_i, y_i) , where $x_i \in \mathbb{R}^n, y_i \in \mathbb{R}, i = 1, 2, \dots, n$, the predicted values can be expressed using K additive functions:

$$y_i = \sum_{k=1}^K f_k(x_i) \quad (3.4.1)$$

In the formula, $x_i, y_i, \hat{y}_i$ are the n -dimensional feature vector of the i -th sample, the actual value of the i -th sample and the predicted value of the i -th sample. The function $f(x)$ maps features to weights of leaf nodes in the tree structure, and can be written as:

$$f(x) = \omega_{q(x)}$$

where ω is the weight of a leaf node, and q is the index of the tree leaf to which the sample is assigned.

The objective function of the XGBoost model consists of a loss function and a regularization term, ensuring model accuracy and generalization ability:

$$Obj = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^n \Omega(f_k) \quad (3.4.2)$$

In this formula:

- Obj is the objective function;
- $l(y_i, \hat{y}_i)$ is the loss function;
- $\Omega(f_k)$ is the regularization term penalizing model complexity.

The model is trained iteratively. In the t -th iteration, the prediction is updated as:

$$y_i^{(t)} = y_i^{(t-1)} + f_t(x_i) \quad (3.4.3)$$

Substituting this into the objective function, we obtain the iterative form:

$$Obj^{(t)} = \sum_{i=1}^n l\left(y_i, y_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t) \quad (3.4.4)$$

Strengths:

- High predictive accuracy due to gradient boosting and regularization;
- Efficient optimization using both gradient and second-order (Hessian) information;
- Robust to missing values and multicollinearity.

Limitations:

- Complex model structure reduces interpretability;
- Requires careful hyperparameter tuning;
- May overfit on small or noisy datasets.

3.4.2 Interpretability with SHAP

In model predictive analysis, understanding the underlying principles and model accuracy is essential. However, for complex models such as ensemble learning algorithms or deep learning models, interpretability is often limited. To address this, the SHAP (SHapley Additive exPlanations) method is employed in this project to visualize and explain the predictions of the XGBoost model.

The SHAP framework decomposes the prediction of each sample into additive feature contributions. Each feature is assigned a marginal contribution value, and the overall predicted value is obtained by summing these contributions:

$$f(x) = E[f(x)] + \sum_{i=1}^n \phi_i \quad (3.4.5)$$

In this formula:

- $f(x)$ is the predicted value of sample x ;
- $E[f(x)]$ is the average prediction;
- ϕ_i represents the SHAP value (marginal contribution) of feature i .

The sign of ϕ_i indicates whether the feature increases or decreases the prediction. This decomposition allows for directional interpretation of feature impact and improves model transparency. Unlike traditional importance measures, SHAP provides fine-grained insight into how each variable influences the model, making it a powerful tool for understanding model behavior [24].

3.5 Model Evaluation Metrics

To assess the performance of the forecasting models in each stage, the following statistical metrics are used in this study:

- **Root Mean Squared Error (RMSE):** RMSE measures the square root of the average of squared differences between predicted and actual values. It penalizes larger errors more heavily and is expressed in the same units as the target variable. The formula is:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}$$

- **Absolute Percentage Error (APE):** APE measures the relative magnitude of the prediction error for a single observation. It is defined as the percentage difference between the actual and predicted value. The formula is:

$$APE_i = \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\%$$

- **Mean Absolute Percentage Error (MAPE):** MAPE calculates the average absolute percentage difference between predicted and actual values, making it useful for comparing models across different scales. The formula is:

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

- **Coefficient of Determination (R^2):** R^2 indicates the proportion of the variance in the dependent variable that is predictable from the independent variables. It ranges from 0 to 1, with higher values indicating better model fit. The formula is:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

These evaluation metrics provide complementary insights: RMSE and MAPE reflect the magnitude and relative size of prediction errors, while R^2 evaluates the model's explanatory power. Together, they help assess the accuracy, robustness, and generalization ability of the hybrid forecasting framework.

CHAPTER 4

Data Description and Exploratory Data Analysis

4.1 Data Sources and Description

This study integrates multiple data sources related to electricity demand forecasting in New South Wales (NSW), with a time span from January 1999 to December 2024. The dataset includes the target variable of total electricity demand, along with several exogenous variables such as meteorological indicators (e.g., temperature, humidity, sunshine, rainfall), socio-economic factors (e.g., population, gross state product), and technology adoption indicators (e.g., electric vehicle registrations and clean energy installations).

All variables were collected from authoritative and publicly available sources to ensure data quality and reproducibility. Specifically:

- Electricity demand and price data are sourced from the Australian Energy Market Operator (AEMO);
- Meteorological variables are obtained from the Bureau of Meteorology (BoM);
- Population and GSP data are from the Australian Bureau of Statistics (ABS);
- Clean energy installation data are compiled from government and energy agency reports;
- Natural gas consumption data are sourced from the Australian Bureau of Energy Statistics (AES).

To ensure consistency across variables, all data were unified to monthly frequency.

4.2 Preprocessing and Frequency Alignment

To align temporal resolution and handle missing values, the following preprocessing strategies were adopted:

- **Electricity demand and price (original frequency: 30-min or 5-min):** Aggregated to monthly averages.
- **Weather variables (temperature, rainfall, humidity, sunshine):** Already monthly; missing values were imputed using the same-month average across all years.
- **Cumulative variables (e.g., solar installations, EV registrations):** Monthly data; missing values filled with 0, assuming no new additions.
- **Quarterly variables (e.g., gas consumption, population):** Uniformly distributed across months within each quarter.
- **Annual variables (e.g., GSP):** Uniformly distributed across months within the corresponding year.

Table 4.1: Summary of Key Variables and Their Sources

Variable Name	Description	Data Source
Total_Demand (MWh)	Electricity consumption	AEMO
Price_Avg (\$/MWh)	Average price of electricity	AEMO
Avg_TAVG_C	Average of temperature	BoM
Avg_TMAX_C	Average of Maximum temperature	BoM
Avg_TMIN_C	Average of Minimum temperature	BoM
Sunshine_MJm ²	Solar exposure	BoM
Precipitation_total	Rainfall	BoM
Population	Population of NSW	ABS
GSP	Gross State Product	ABS
Cumulative_total_systems	Clean energy device installations like solar panel or rooftop PV	Government/Energy agencies
Avg_Humidity	Average of humidity	BoM
Avg_Max_Humidity	Average of Maximum humidity	BoM
Avg_Min_Humidity	Average of Minimum humidity	BoM
Gas_Consumption	Natural gas usage	AES
EV_registration_number	Electric vehicle	ABS / Transport departments

4.3 Variable Overview

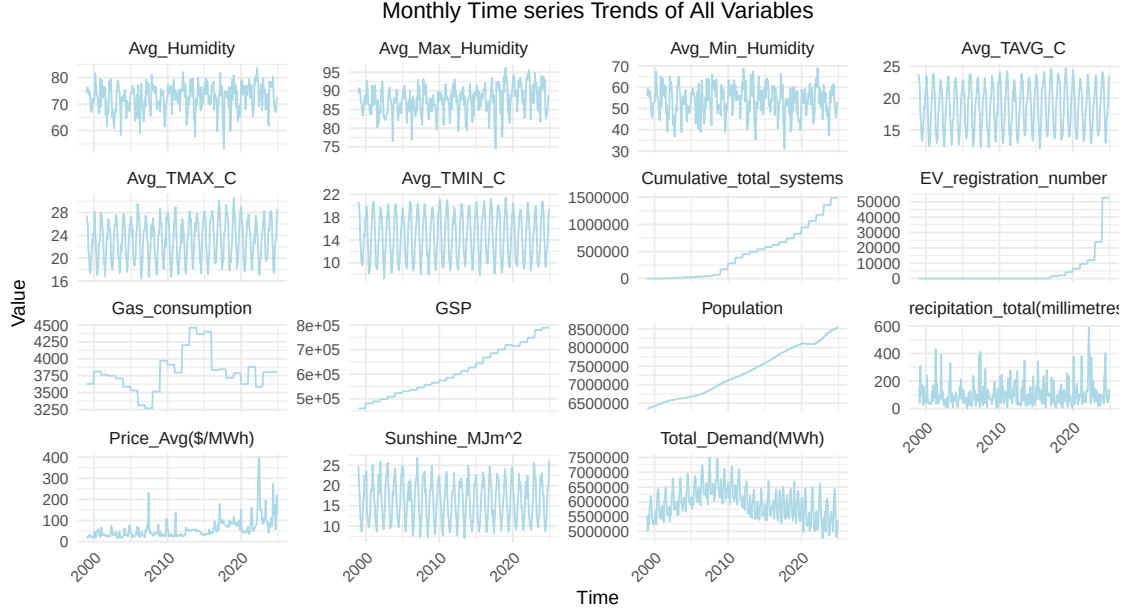


Figure 4.1: Time Series Trends of All Key Variables (1999-2024)

Note. This figure presents the monthly time series trends of all key variables used in the forecasting models over the period 1999–2024.

Figure 4.1 displays the monthly time series trends of all variables used in this study from 1999 to 2024. Notably, temperature-related variables (for example the average temperature) exhibit clear seasonal patterns. Electricity demand and solar exposure also demonstrate cyclical fluctuations. In contrast, variables such as population, cumulative solar device installations, and GSP (Gross State Product) show a steady upward trend, reflecting socio-economic and technological development. Some variables (e.g., precipitation, price) show greater volatility over time.

Figures 4.2 and 4.3 present the distribution of key variables through histograms with overlaid density curves.

The distribution of total electricity demand appears relatively symmetric and roughly follows a normal distribution, with most observations concentrated around 6.0 million MWh.

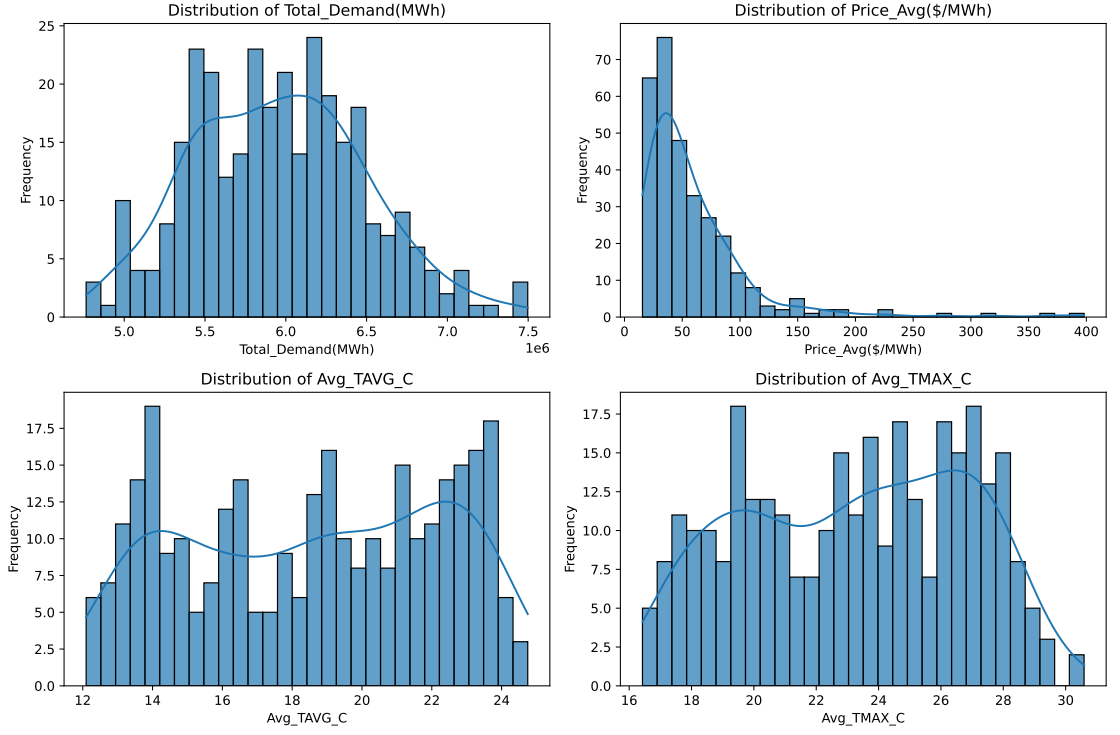


Figure 4.2: Histogram and KDE of Key Features

In contrast, the distribution of average electricity price is highly right-skewed, indicating that while prices are generally low in most months, a few months experienced exceptionally high prices.

The distributions of average mean temperature, average maximum temperature, average minimum temperature, and sunshine duration exhibit noticeable multimodal patterns. These fluctuations likely reflect seasonal variations in weather and solar exposure.

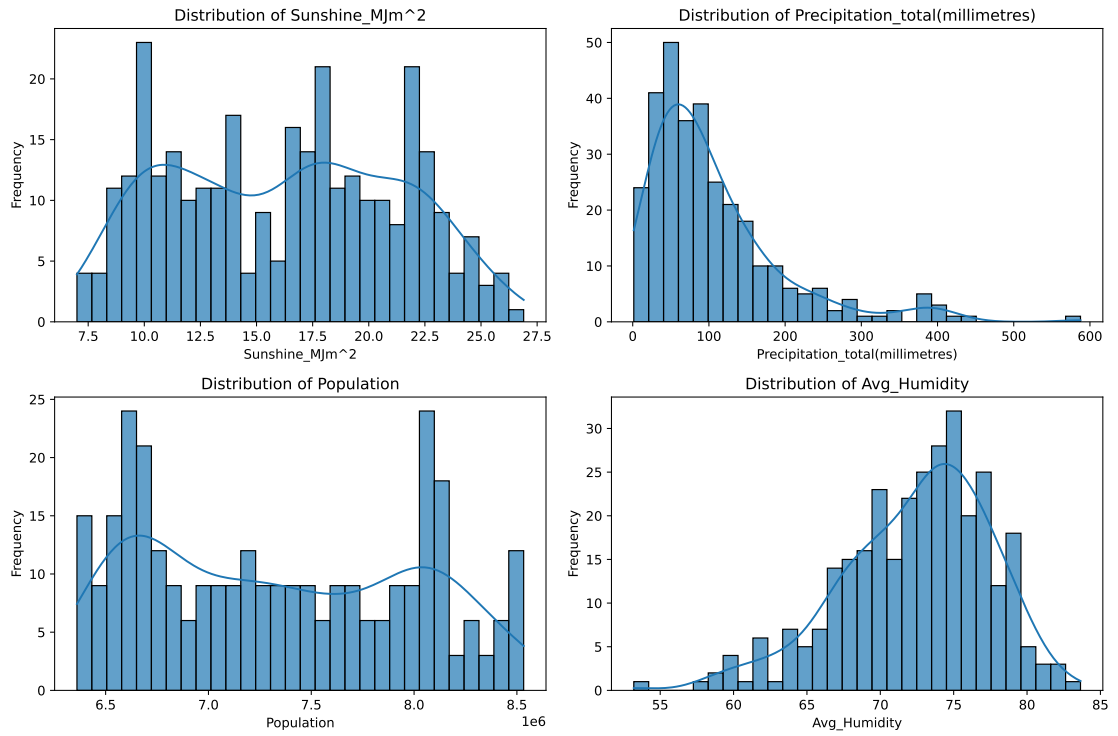


Figure 4.3: Continued Distribution Plots of Key Variables

Note. This figure and above illustrates the distribution and kernel density estimates (KDE) of selected key variables. These plots help reveal skewness, peak value, and overall shape of the variables

Total monthly precipitation is predominantly low, though the distribution is skewed to the right due to a few months with unusually heavy rainfall.

Lastly, average relative humidity shows a near-normal distribution with a slight left skew, suggesting that higher humidity levels are more common, whereas low-humidity conditions occur less frequently.

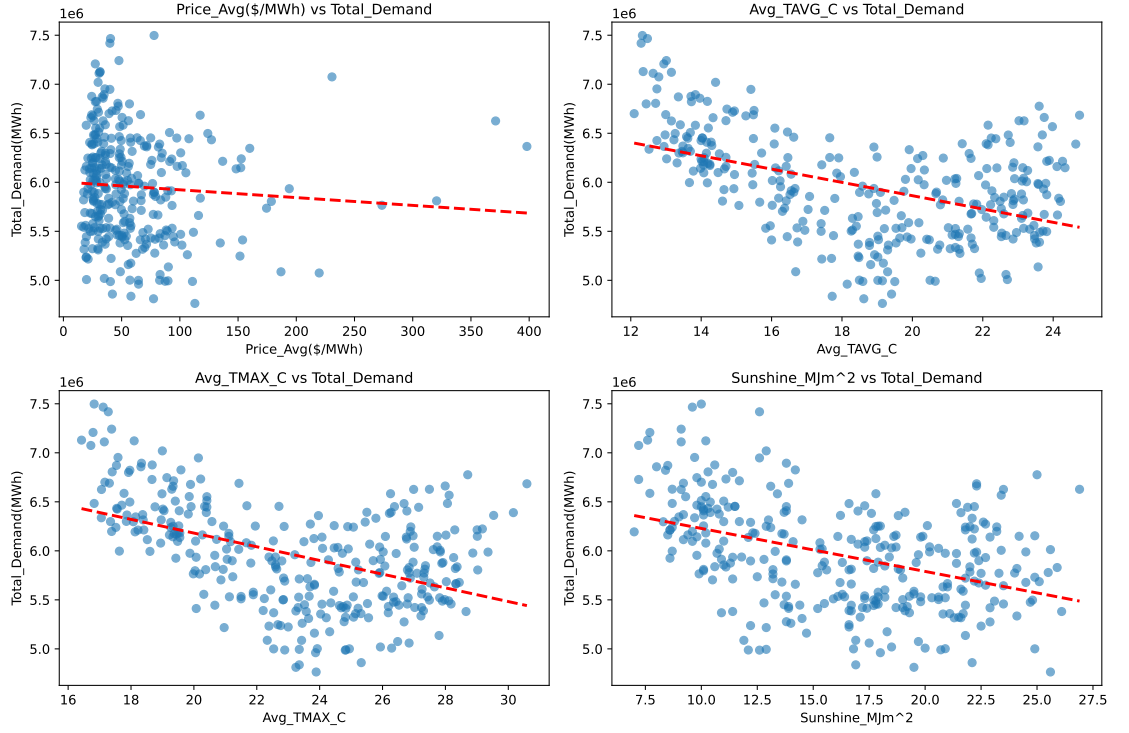


Figure 4.4: Scatter Plots with Trend Lines between Predictors and Total Demand

Figures 4.4 and 4.5 present the scatter plots of the relationships between each explanatory variable and the total electricity demand with trend regression lines to assist in observation.

It can be seen from the graph that the average, average maximum, average minimum temperature and the amount of sunshine show a certain negative correlation trend with the total electricity demand, indicating that in months with higher temperatures or stronger sunshine, the electricity demand may slightly decrease. This relationship may be related to the saturation or substitution behavior of the refrigeration load within a certain range.

In contrast, the relationship between variables such as precipitation, humidity and population and total electricity demand is not very obvious, and the trend line is relatively flat, indicating that the influence of these factors on electricity demand may be more indirect or nonlinear. In addition, the relationship between average electricity prices and electricity demand is also relatively weak, which may be affected by the lagging response of the price mechanism and electricity consumption behavior.

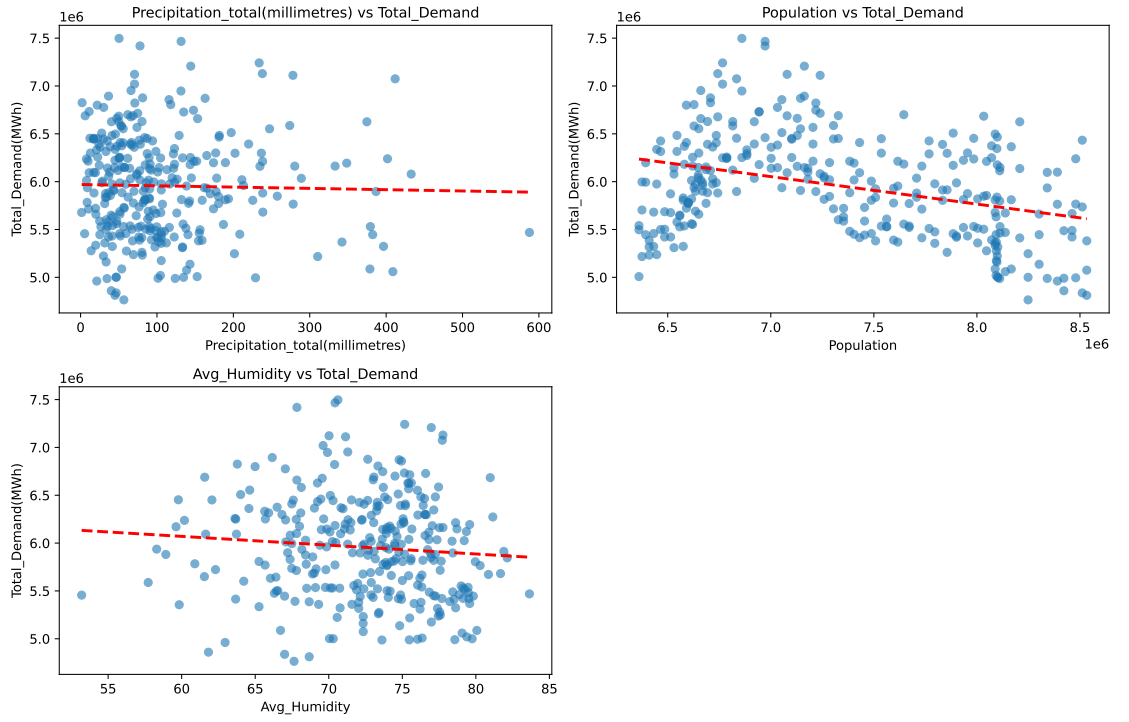


Figure 4.5: Continued Scatter Plots for Remaining Variables vs. Total Demand

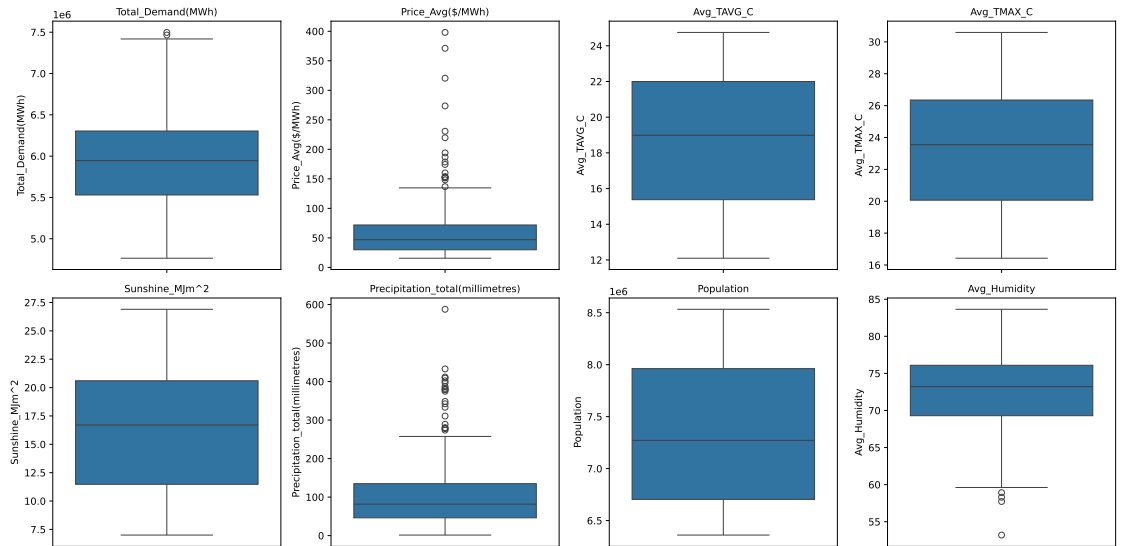


Figure 4.6: Boxplots of Key Variables

Figure 4.6 shows the box plots of all the main variables, which are used to reflect the distribution range, central tendency, and whether there are outliers of the variables. It can be seen from the graph that there are obvious outliers in the average electricity price and precipitation, indicating that there may be extremely high prices or heavy rainfall in some months. The distribution of variables such as temperature, humidity and sunlight is relatively concentrated, indicating that

their fluctuations are relatively stable. The distribution of total electricity demand is approximately symmetrical, further confirming its characteristic of being close to a normal distribution.

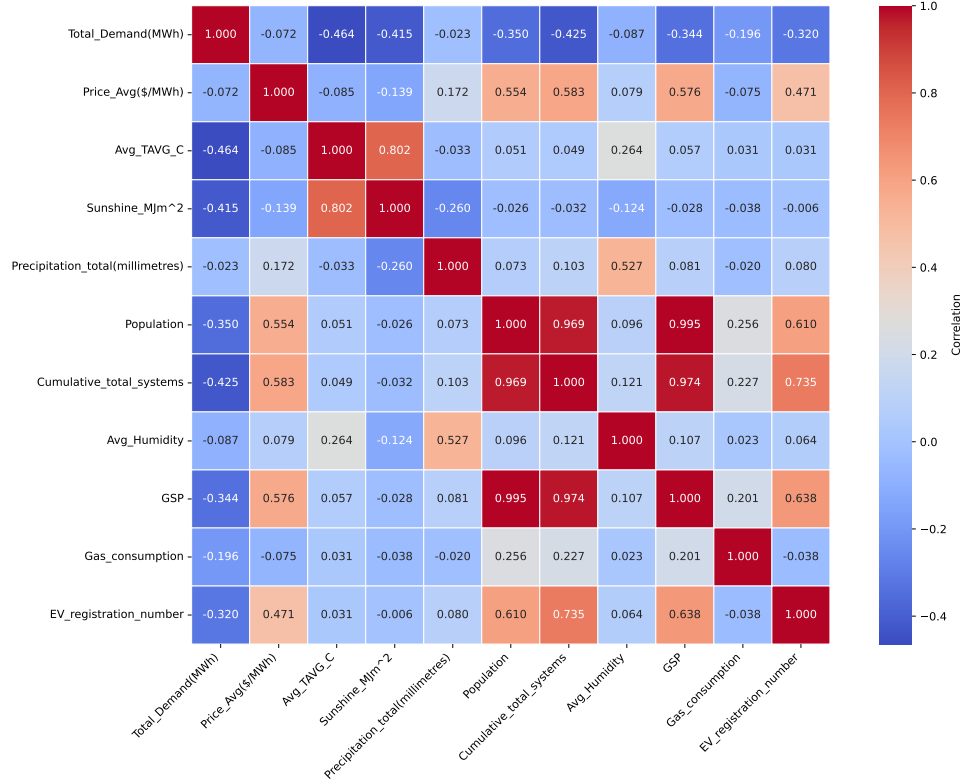


Figure 4.7: Correlation Heatmap

Figure 4.7 shows the Pearson correlation coefficient heatmap among the main variables. It can be seen that the cumulative installed capacity of solar energy systems is highly positively correlated with the population (0.969), and both are also strongly correlated with the state gross domestic product, indicating that there may be a certain degree of collinearity among these indicators. Electricity demand is moderately negatively correlated with multiple meteorological variables (such as average temperature, average maximum temperature, and sunshine duration), further verifying the trend in the previous scatter plot. However, its correlation with prices is relatively weak (-0.072), and it may be affected by policies factors. In addition, there is a clear positive correlation between the number of registered electric vehicles and the population as well as the installed capacity of photovoltaic systems, which is in line with the growth trend of renewable energy and the electrification of transportation in reality

4.4 Training and Testing Split

To evaluate model performance, the dataset is divided into training and testing sets. Data from January 1999 to December 2022 are used for model training and parameter tuning, while the final 24 months (January 2023 to December 2024)

are reserved for out-of-sample testing. This split enables the assessment of model generalization ability and forecasting accuracy on unseen future data.

CHAPTER 5

Results and Discussion

5.1 Stage One: Trend Extraction Using Prophet

To evaluate the model's ability to extract long-term structural trends, the Prophet model was trained on the training set and used to generate trend forecasts on both training and testing periods. The predicted trend values are compared with actual electricity demand to assess the trend component visually. Additionally, after removing the estimated trend component, we applied the Augmented Dickey-Fuller (ADF) test and rolling mean/standard deviation plots to assess the stationarity of the residual series. This validation step ensures that the remaining seasonal component is stable and suitable for SARIMA modeling in the next stage.

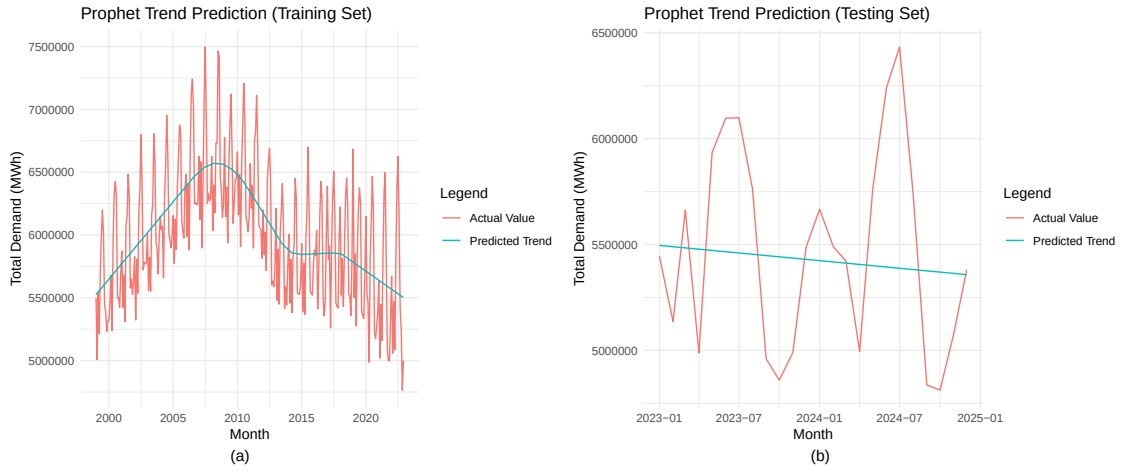


Figure 5.1: $\hat{y}_{trend}$ vs y

Figure 5.1 displays the predicted trend components generated by the Prophet model versus the actual electricity demand. In subfigure (a), the model is fitted on the training data from 1999 to 2022. The red line in the figure represents the actual monthly electricity demand, while the blue line indicates the long-term trend extracted by the Prophet model. It can be seen that this model effectively captured the obvious upward trend between 2000 and 2010, and then the trend stabilized or even slightly declined, reflecting the macro structural changes in electricity demand.

In subfigure (b), the trend prediction results of the model on the test set (2023–2024) are presented. Despite the fact that the actual demand during the testing period fluctuated sharply due to seasonal and external factors, the trend

predicted by the Prophet model still maintained a stable downward trend, indicating that the model has good generalization ability and can extract potential structural signals from new data.

Table 5.1: Indicators Values of $\hat{y}_{trend}$ vs y

Dataset	RMSE (MWh)	MAPE (%)
Training set	399625.2	5.63
Test set	468883.7	7.2

Note. This table presents the evaluation metrics for the model’s fit to the trend component $\hat{y}_{trend}$. It serves as the first stage in the three stages forecasting framework, highlighting how each stage incrementally improves prediction performance.

Although cross-validation (CV) is commonly used to assess the generalization ability of predictive models, it is not strictly required in the first stage of this study where the Prophet model is employed. The reason is that Prophet is not used here for forecasting future values but rather as a decomposition tool to extract the long-term trend component from the electricity demand time series. Its role is descriptive, aiming to separate the original series into interpretable components (trend, seasonality, and residuals), rather than predictive.

Nevertheless, to evaluate the quality of the trend extracted by Prophet, we calculate the RMSE and MAPE between the predicted trend and the actual total electricity demand. On the training set, RMSE is 399,625.2 and MAPE is 5.63%, while on the test set, RMSE reaches 468,883.7 with a MAPE of 7.15%. These results demonstrate that the model effectively captures the underlying trend and maintains good generalization performance even on unseen data.

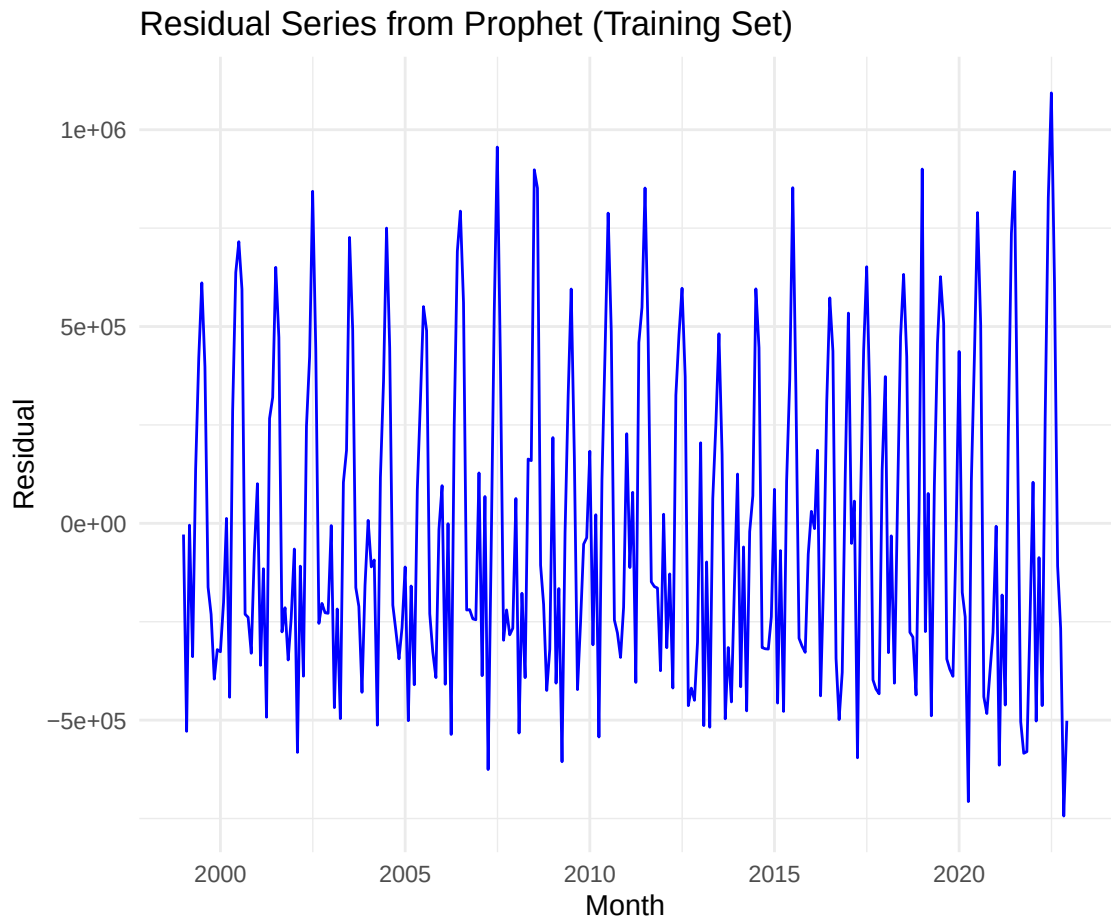


Figure 5.2: Time Series Data of Residual

To verify the validity of the trend extracted by Prophet, the residual series calculated as the difference between the actual values and the predicted trend was visualized (Figure 5.2). The time series of residuals shows significant periodic fluctuations, indicating the presence of strong seasonality components that have not been captured by the trend model.

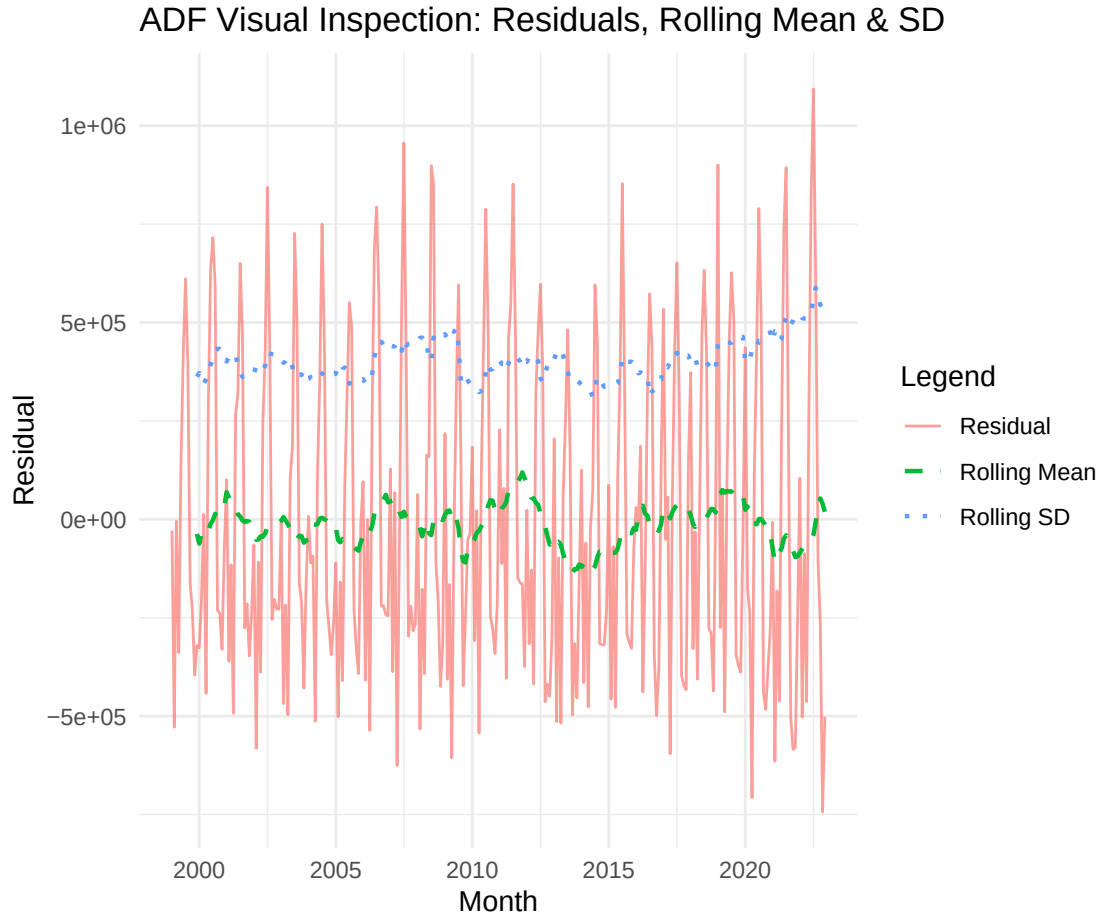


Figure 5.3: Residual with Rolling Mean and Rolling SD

Note. This plot shows the residual series along with its rolling mean and rolling standard deviation. The relatively constant mean and variance over time visually indicate that the residuals are stationary, providing theoretical support for applying SARIMA in the next stage of forecasting.

To further verify the stationarity of the residual series, we plot the rolling mean and rolling standard deviation (rolling SD) (Figure 5.3). These two statistics are helpful in determining whether the mean and variance of a sequence remain stable at different time windows, which is a basic assumption that a stationary sequence needs to satisfy. The plot has shown that both rolling mean and rolling SD remain relatively stable over time, with no significant trends or fluctuations.

Combined with the Augmented Dickey-Fuller (ADF) test result (p -value = 0.01), this provides strong evidence that the residuals are stationary. It confirms that the series is suitable for seasonal modeling in the next stage. Since the second stage of SARIMA relies on the data remaining after removing the trend from the first stage to model seasonality, if the data is not stable and SARIMA is still used for modeling, it would violate SARIMA's basic assumption (which requires the input to be a stationary sequence), resulting in poor fitting of the SARIMA model, false periodic components or overfitting. The periodicity of the data cannot be extracted, thus affecting the rationality of the residual modeling in the third stage.

5.2 Stage Two: Seasonal Component Modeling with SARIMA

After completing the trend modeling in the first stage, we noticed that the remaining part (that is, the difference between the actual power load and the long-term trend) still showed obvious seasonal fluctuations. we introduced the Seasonal Autoregressive Integral Moving Average Model (SARIMA) in the second stage to capture this seasonal component.

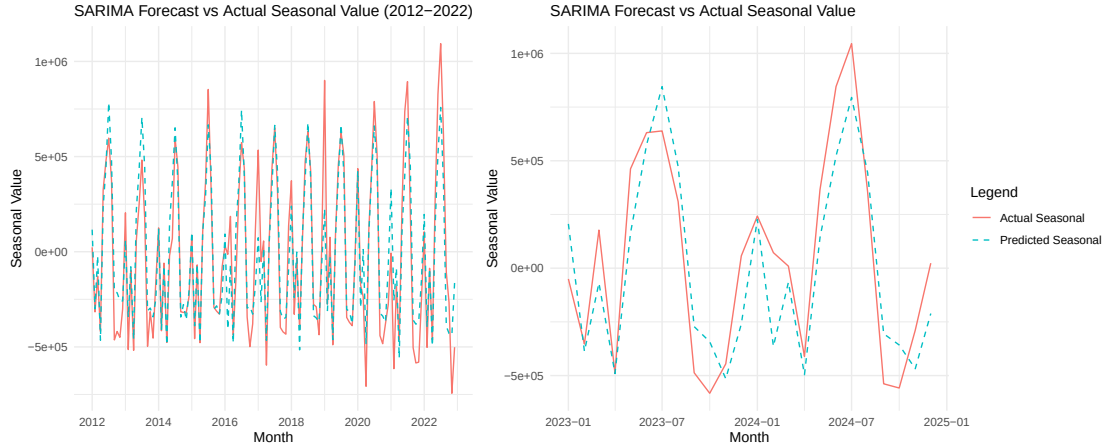


Figure 5.4: SARIMA $\hat{y}_{\text{seasonal}}$ vs $y - \hat{y}_{\text{trend}}$

Figure 5.4 illustrates the performance of the SARIMA model in capturing seasonal components for both the training set and the test set. In the training period (2012–2022, only ten years are shown due to format of plots), the predicted seasonal values closely track the actual seasonal fluctuations, with only minor deviations near peak points. In the test period (2023–2024), the model continues to capture the main seasonal patterns, although some phase lag and amplitude discrepancies are evident, especially around changepoints.

Table 5.2: Indicators Values of $\hat{y}_{\text{seasonal}}$ vs $y - \hat{y}_{\text{trend}}$

Dataset	RMSE (MWh)	MAPE (%)
Training set	146749.4	137.2
Test set	214933.9	175.6

Note. This table presents the goodness-of-fit metrics of the SARIMA model in capturing the seasonal component.

The RMSE values indicate in Table 5.8 indicate a moderate increase in forecasting error on unseen data. Meanwhile, the MAPE are 137.2% and 175.6% respectively. Although the actual seasonal values are not small, the MAPE still exceeds 1. This is mainly due to the following reasons:

- The SARIMA model was deliberately constrained to avoid fitting non-seasonal structures. As a result, the model refrains from capturing atypical fluctuations or complex structures, leading to larger residuals at certain points.
- Seasonal sequences have the characteristic of positive and negative fluctuations. When the predicted direction is opposite to the true value direction, although the absolute error is limited, the relative error is significantly magnified, thereby increasing the MAPE index.
- When performing recursive predictions on the test set, the model may have periodic phase shifts or amplitude scaling errors, further intensifying the deviation of the predictions in a relative sense.

5.2.1 Residual Diagnostics of SARIMA Model via ACF

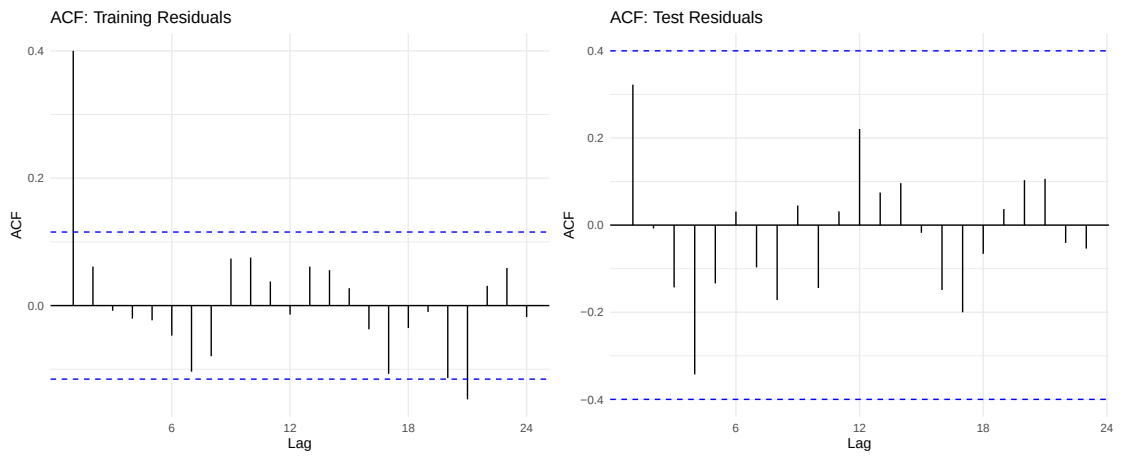


Figure 5.5: ACF of Training and test set

Figure 5.4 shows the residual Autocorrelation Function (ACF) graph of the SARIMA model on the training set and test set are presented. It can be seen that for the training set, except for lag=1 which is slightly higher and close to the significance limit, the autocorrelation coefficients of the other lag orders are basically within the confidence interval and do not show significant autocorrelation. This indicates that the model effectively fits the major seasonal and structural fluctuations on the training data, and its residuals are approximately white noise, which conforms to the assumption of residual independence of the model. For the test set, compared with the training set, although the residuals of the test set show more obvious fluctuations at some lag points (such as lag= 4, 12, 17), they still do not cross the significance boundary of the 95% confidence interval overall, indicating that the model does not have systematic periodic biases when making recursive predictions. In summary, the residual diagnostics from both training and test sets show no significant autocorrelation, supporting the conclusion that the SARIMA model is adequately specified and generalizes reasonably well for seasonal component forecasting.

5.2.2 Interpretation of RMSE Comparison between SARIMA and Hybrid Forecasts

It is important to note that comparing SARIMA's predicted seasonal component with $y - \hat{y}_{\text{trend}}$ serves only to assess the model's ability to capture seasonality, not the overall forecasting performance. Since the trend extracted by Prophet contains estimation errors, the residual $y - \hat{y}_{\text{trend}}$ does not represent the true seasonal component. Therefore, the RMSEs and MAPE values from these two evaluations are not directly comparable.

5.2.3 Applied Extracting Trend and Forecasting Seasonal

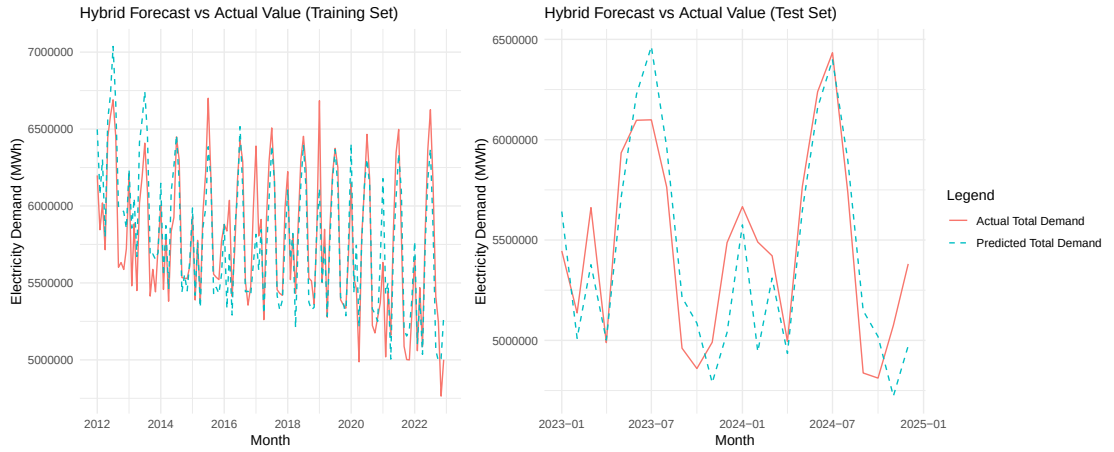


Figure 5.6: $\hat{y}_{\text{seasonal}} + \hat{y}_{\text{trend}}$ vs y Total Electricity Demand

Table 5.3: Indicators Values of $\hat{y}_{\text{seasonal}} + \hat{y}_{\text{trend}}$ vs y

Dataset	RMSE (MWh)	MAPE (%)
Training set	175665.9	2.15
Test set	252619.9	3.97

Note. This table presents the goodness-of-fit metrics for the combined predictions of the trend and seasonal components.

Figure 5.6 shows the fitting effect of the Hybrid Forecast obtained by superimposing the trend components output by the Prophet model in the first stage and the seasonal components predicted by the SARIMA model in the second stage on the electricity demand. From the training set, the predicted values are highly consistent with the actual trend of electricity demand, successfully capturing the long-term downward trend and monthly seasonal fluctuations. In terms of the indicators, the RMSE on the training set is 175,665.9 and the MAPE is 2.15%, indicating that the model has a relatively high fitting accuracy on the known data. In the test set

section, despite individual peak deviations (such as underestimation of predictions for mid-2023 and the end of 2024), the overall trend and seasonal structure remain highly consistent with the actual values. The RMSE of the test set is 252,619.9, and the MAPE is 3.97%. Although slightly higher than that of the training set, it is still at a relatively low level, indicating that the model has good generalization ability. The ACF test results combined with the aforementioned residuals show that the SARIMA model does not leave significant autocorrelation in the residuals, further verifying the effectiveness of this hybrid modeling scheme in structural modeling. Therefore, we believe that the two-stage modeling method based on Prophet + SARIMA can accurately reconstruction the long-term trend and seasonal components of electricity demand without introducing the risk of complex overfitting, providing a good foundation for subsequent residual modeling.

5.3 Stage Three: Residual Forecast using XGBoost

5.3.1 *XGBoost Hyperparameter Tuning Strategy*

To improve the performance of the XGBoost model, a sequential tuning strategy was adopted in this study. First, some parameters that have a relatively small impact on the model performance, such as `subsample`, `colsample_bytree`, `lambda` and `alpha`, were fixed to narrow the search space. Subsequently, the focus was on adjusting `eta` (learning rate) and `max_depth` (tree depth), which have the greatest impact on model complexity and learning rate. After finding the optimal combination through cross-validation, further fine-tune other regularization and sampling-related parameters. This method can ensure the efficiency of parameter adjustment while avoiding overfitting of the model and enhancing the generalization ability.

Table 5.4: Forecasting Performance under Different Learning Rates and Tree Depths

Best Iteration	ETA	Max Depth	CV-Mean test RMSE	Train RMSE (MWh)	Test RMSE (MWh)
28	0.5	1	125845.6	108842.6	143100.9
36	0.5	3	120354.1	55624.36	152635.3
72	0.5	5	123818.7	7847.129	210028.2
157	0.1	1	124397	105636.6	145191.6
101	0.1	3	118998.2	72336.53	163365.8
103	0.1	5	118115	46105.76	167283.7
206	0.05	1	125660	111057.2	149518.1
201	0.05	3	118222.1	72899.27	162144.4
1737	0.005	1	126134.9	113277.9	151667.3
1773	0.005	3	118255.1	76415.83	163392.5

Note. This table summarizes the tuning results for the XGBoost model using different learning rates (ETA) and maximum tree depths (Max Depth).

In the first stage of hyperparameter tuning, keeping other hyperparameters fixed (`subsample = 0.7`), `colsample_bytree = 0.7`, `lambda = 10`, `alpha = 5`). Here, `subsample` controls the fraction of training instances randomly sampled for each boosting round, which helps prevent overfitting. `colsample_bytree` specifies the fraction of features (columns) to be randomly sampled for each tree, introducing additional randomness. `lambda` is the L2 regularization term on leaf weights, which penalizes large coefficients and encourages model generalization, while `alpha` is the L1 regularization term that can induce sparsity in the model.

As shown in Table 5.4, the combination of `eta = 0.5` and `max_depth = 1` achieved the lowest test RMSE (143100.9), indicating strong predictive accuracy. However, models with lower learning rates, such as `eta = 0.1`, produced very similar performance (e.g., RMSE = 145191.6) while offering more stable convergence. This suggests that a relatively shallow tree (depth = 1 or 3) and moderate learning rate (e.g., 0.1) may balance predictive performance with generalization ability. Notably, deeper trees (e.g., `max_depth = 5`) led to substantial overfitting, as evidenced by low train RMSE and significantly larger test RMSE (e.g., 210028.2 when `eta = 0.5`, `max_depth = 5`).

So the second part of parameter tuning that follows will be carried out between `colsample_bytree` and `subsample`.

Table 5.5: Forecasting Performance under Different Subsample and Colsample

Best Iteration	Subsample	Colsample Bytree	CV-Mean test RMSE	Train RMSE (MWh)	Test RMSE (MWh)
122	0.5	0.5	125904.6	110574.7	152410.5
122	0.5	0.6	124877.3	110313.1	150528.6
122	0.5	0.7	125664.8	110129.2	149379.5
122	0.5	0.8	125646.5	109688.9	150698.4
158	0.6	0.5	124694.5	106274.5	151590.7
147	0.6	0.6	124773.4	107263.9	151179
158	0.6	0.7	124849.8	105884.3	152991.8
147	0.6	0.8	125121.1	106669.5	148952.6
147	0.6	0.9	124609.1	106628.1	149495.9
157	0.7	0.5	124540.7	106083.6	148117.8
161	0.7	0.6	124942.8	106189.8	147779.7
157	0.7	0.7	124397	105636.6	145191.6
157	0.7	0.8	124796.4	105616.7	143333
164	0.7	0.9	124795.5	104505.8	144171.2

Note. This table presents the parameter tuning results of the XGBoost model under different combinations of subsample ratios (`subsample`) and feature sampling ratios (`colsample_bytree`).

After determining `eta = 0.1` and `max_depth = 1`, the model was tuned on combinations of `subsample` and `colsample_bytree`. As shown in Table 5.5, the lowest test RMSE (**143333**) was achieved when `subsample = 0.7` and `colsample_bytree = 0.8`, indicating this combination helps reduce overfitting while maintaining good generalization.

Then, with the above values fixed, the tuning continued for regularization-related parameters: `gamma` and `min_child_weight`. According to Table 5.6, the combination of `gamma = 0` and `min_child_weight = 4` yielded the best test RMSE of **143246.4**, showing that less aggressive pruning and a moderate minimum child weight are better suited to the data in this case.

Table 5.6: Forecasting Performance under Different Gamma and Min Child Weight

Best Iteration	Gamma	Min Child Weight	CV-Mean test RMSE	Train RMSE (MWh)	Test RMSE (MWh)
157	0	1	124796.4	105616.7	143333
157	0	3	124697.3	105616.7	143333
157	0	4	124552	105648.8	143246.4
178	0	5	124787.6	104516.1	147731.1
157	0.5	1	124796.4	105616.7	143333
157	0.5	3	124697.3	105616.7	143333
178	0.5	5	124787.6	104516.1	147731.1
178	0.5	7	124662.9	104587.9	147800.7
157	0.7	1	124796.4	105616.7	143333
178	0.7	5	124787.6	104516.1	147731.1
178	0.7	7	124662.9	104587.9	147800.7

Note. This table presents the XGBoost tuning results under different values of `gamma` and `min_child_weight`. `Gamma` controls the minimum loss reduction required for further partitioning a tree leaf, while `min_child_weight` specifies the minimum sum of instance weights needed in a child node. These parameters help prevent overfitting and improve model generalization during residual prediction.

The final chosen parameter set was validated via 10-fold cross-validation, as illustrated in Table 5.7 and Figure 5.7, where the best iteration number was 157 and the minimum CV RMSE was 124552.

Table 5.7: Final hyperparameters

ETA	Max Depth	Subsample	Colsample Bytree	Gamma	Min Child Weight
0.1	1	0.7	0.8	0	4

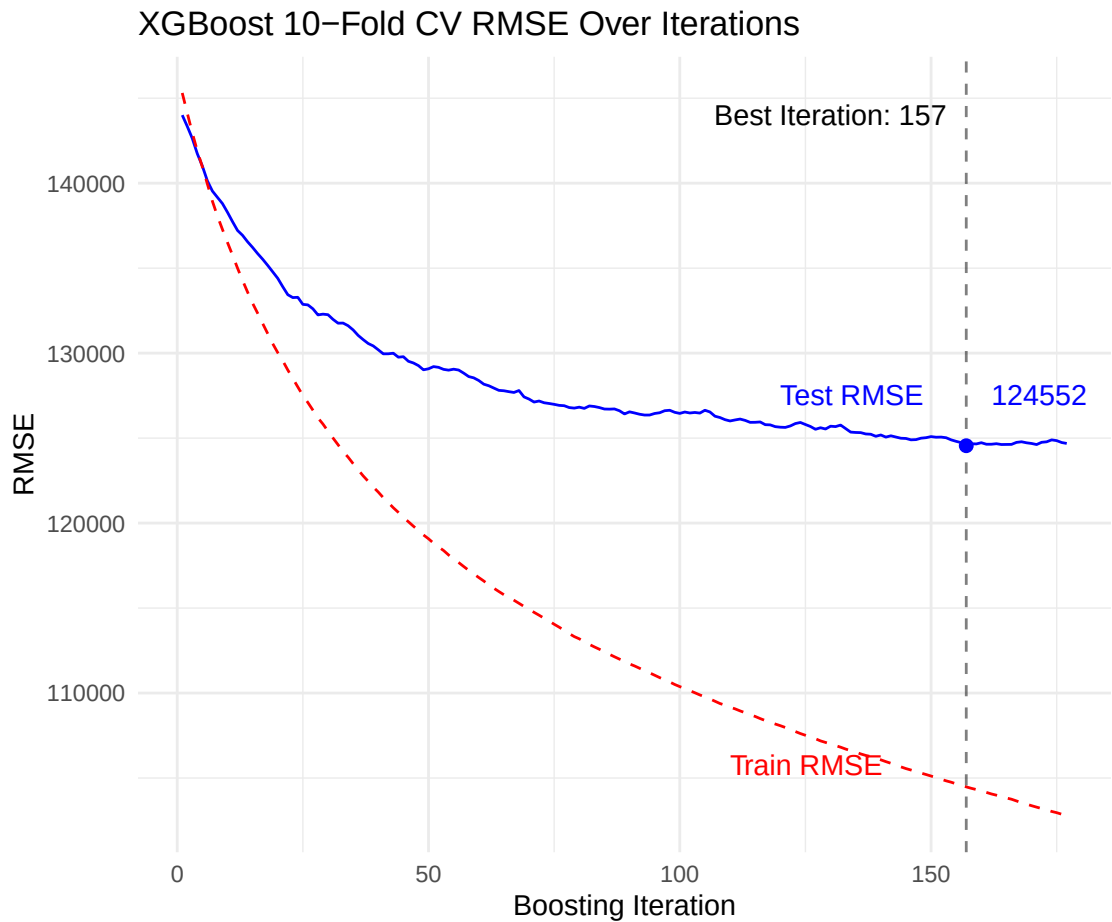


Figure 5.7: 10-Fold Cross-Validation RMSE Over Iterations

5.3.2 Residual Modeling and Final Forecast Evaluation

Figure 5.8 shows the performance of the XGBoost model in fitting and predicting the residuals on the training and test sets, respectively. In the training set, the fitted residuals generally follow the actual residuals, particularly in periods with moderate fluctuations. This indicates that the model has captured a substantial portion of the nonlinear patterns remaining after trend and seasonal components were removed. However, discrepancies remain, especially during periods with extreme spikes in residuals, which results in a training set R^2 of 0.4988 and RMSE of 105648.8.

This suggests that the model explains approximately 49.9% of the variance in the residuals.

On the test set, the model continues to track the general direction of the actual residuals, although some discrepancies can be observed at specific peaks and troughs. Despite these variations, the test set R^2 reaches 0.5754 with an RMSE of 143246.4, slightly outperforming the training set in terms of explanatory power. This result suggests that the model has not overfitted the training data and generalizes reasonably well to unseen data.

Table 5.8: Indicators Values of $\hat{y}_{\text{residual}}$ vs $y - \hat{y}_{\text{trend}} - \hat{y}_{\text{seasonal}}$

Dataset	RMSE (MWh)	R^2
Training set	105648.8	0.4988
Test set	143246.4	0.5754

Note. This table reports the performance of the XGBoost model in predicting the residual component $\hat{y}_{\text{residual}}$, evaluated against the unexplained part $y - \hat{y}_{\text{trend}} - \hat{y}_{\text{seasonal}}$. RMSE and R^2 indicate the model's accuracy and explanatory power in the final stage of the hybrid framework.

In summary, the XGBoost model effectively models the residuals from the first two stages. It corrects remaining errors that were not captured by the trend and seasonal components. Furthermore, It is worth noting that the residual values modeled in Stage 3 are substantially smaller in magnitude compared to the original demand series, and may fluctuate around zero. As a result, MAPE becomes sensitive to small denominators, often leading to inflated error percentages. Therefore, RMSE and R^2 are preferred for evaluating model performance in this residual modeling stage.

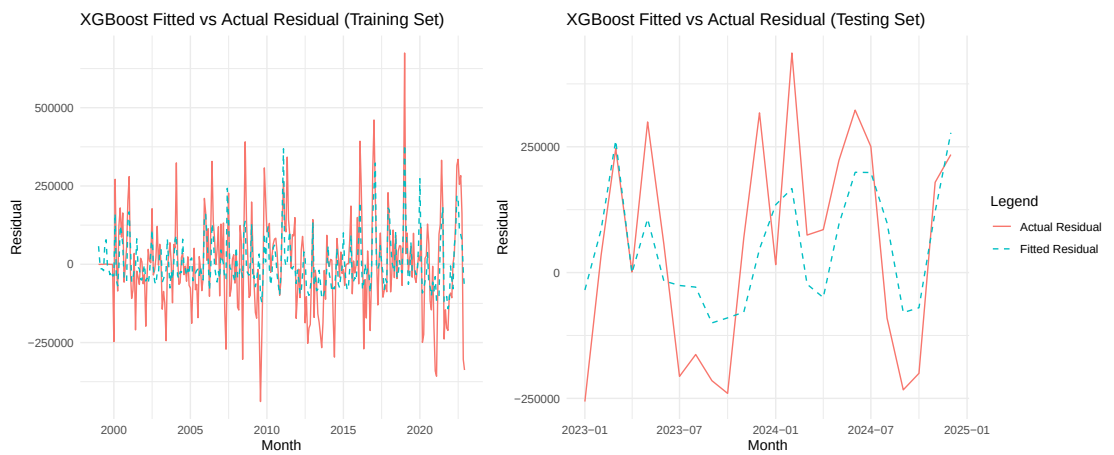


Figure 5.8: Forecast vs Actual Residuals in Training and Test set

5.4 Final Hybrid Forecast Evaluation

Figure 5.9 and Figure 5.10 display the comparison between actual electricity demand and the fitted values generated by the final three-stage hybrid model on the training set and test set, respectively.

As shown in Figure 5.9, the fitted values closely follow the actual demand throughout the entire training period, capturing both the long-term trend and the seasonal fluctuations effectively. The prediction line almost overlaps with the actual values, which indicates that the model has learned the underlying patterns well during the training process.

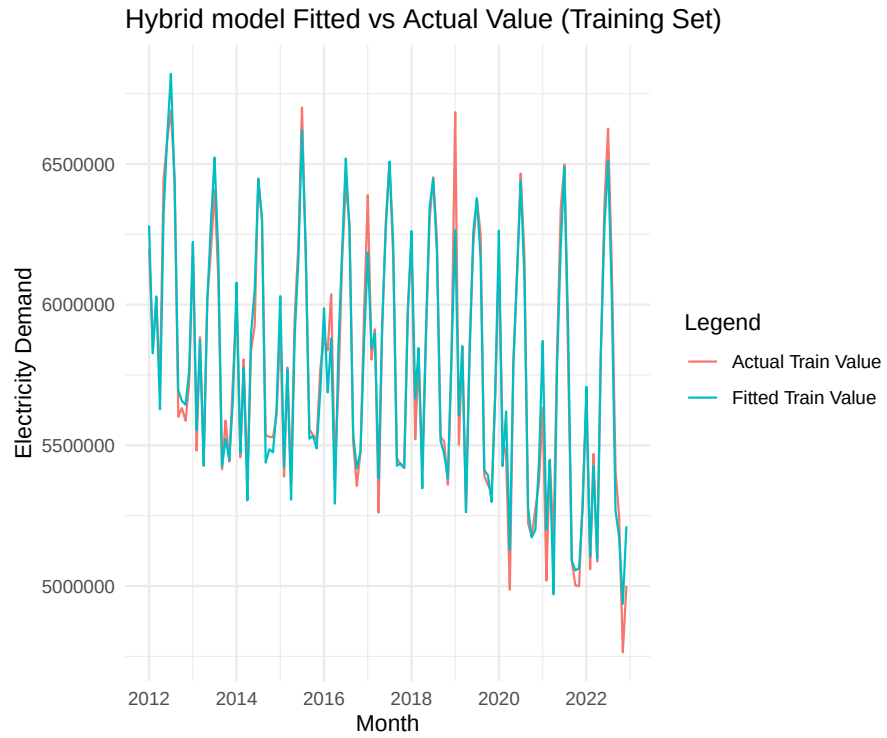


Figure 5.9: Final Training Fitted Values vs Actual

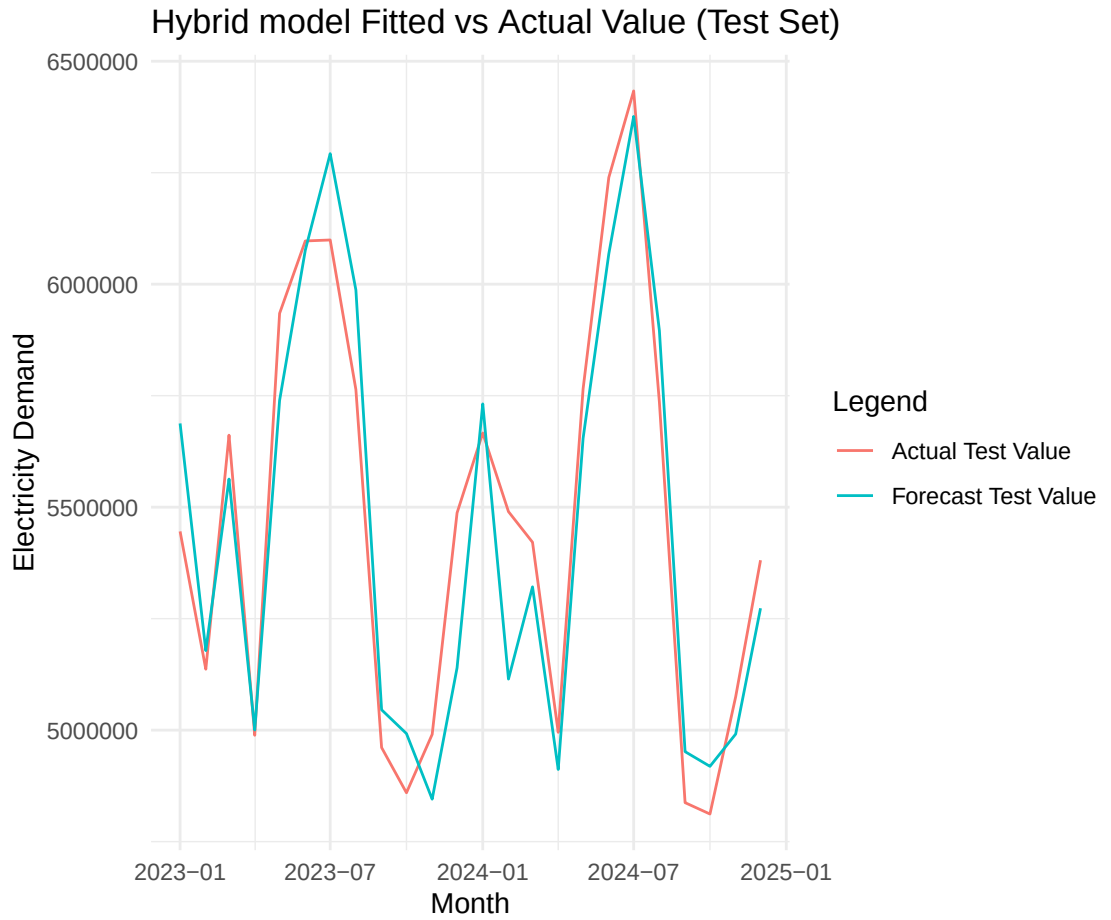


Figure 5.10: Final Test Forecast Values vs Actual

In general, the visual comparison in both the training and testing sets demonstrates that the hybrid framework has successfully integrated the advantages of each modeling stage: trend extraction (Prophet), seasonality modeling (SARIMA), and residual correction (XGBoost). These results provide compelling evidence that the hybrid approach yields robust and accurate forecasts for monthly electricity demand.

5.4.1 Model Interpretation with SHAP

Figure 5.11 shows the SHAP summary plot for the XGBoost residual model. In general, the variables influencing the residual of electricity demand can be classified into three major categories: (i) meteorological conditions (temperatures and humidity), (ii) market signals (electricity price, gas consumption), and (iii) demographic factors (population).

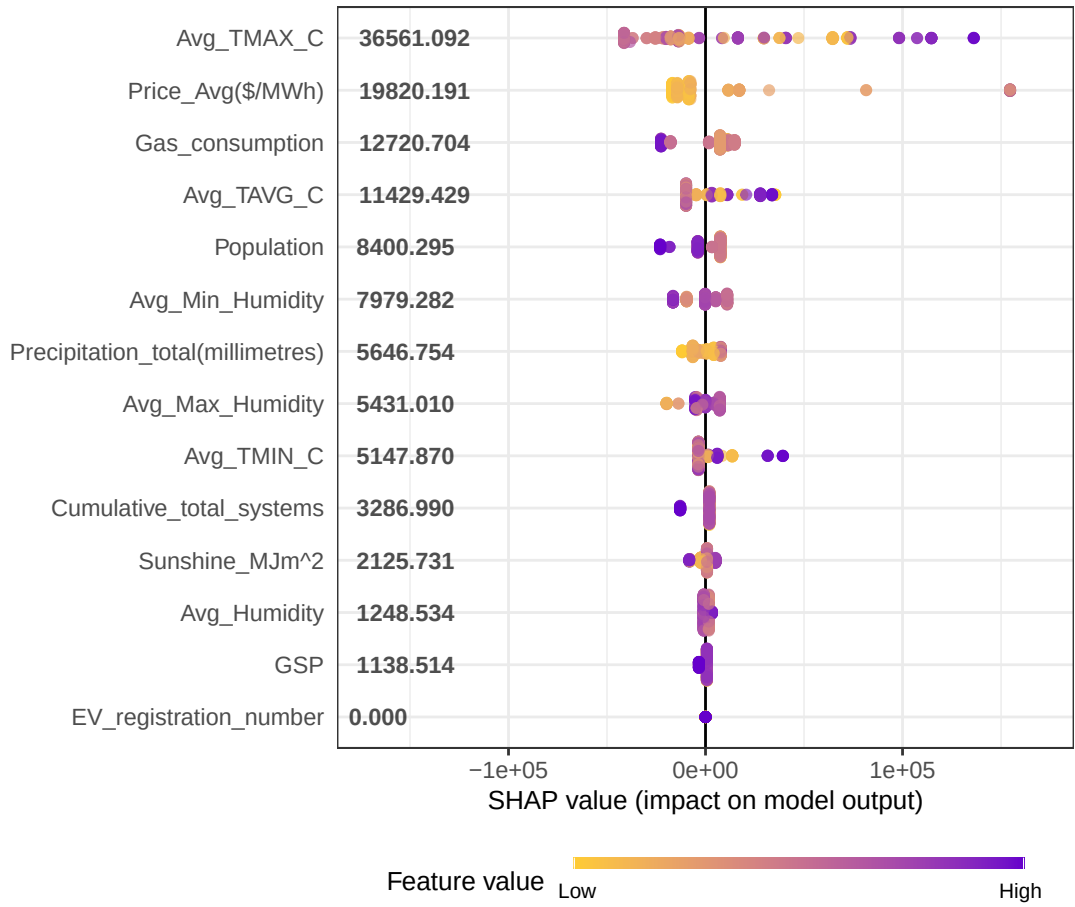


Figure 5.11: SHAP Summary Plot

High-impact features. The average maximum temperature is the most influential variable, with an average absolute SHAP value of 36,561. Higher peak temperatures usually push up the residual values of the model, indicating that in extreme high-temperature weather, the electricity demand brought by cooling equipment such as air conditioners has not been fully captured by the first two stages (trend + seasonality). This is also consistent with practical experience: the hotter the weather, the higher the demand for electricity, especially during the midday and afternoon periods in summer.

The two variables that closely follow are the average price of electricity (19,820) and the consumption of natural gas (12,721). High electricity prices may curb nonessential electricity demand or reflect the price increase caused by the increase in system load. The consumption of natural gas is often related to industrial production or heating demands, and these factors also bring about additional electricity usage, thereby affecting the model residuals.

Medium-impact features. The SHAP values of the average temperature and the average minimum temperature are both around 11,000, and they jointly characterize the changes in the base demand for electricity. The population size (8,400) reflects the part of Australia's long-term electricity demand growth trend

that has not been fully modeled by the Prophet trend term, and thus still has a certain impact on the residuals.

Low-impact features. Variables, such as solar exposure, the state gross domestic product (GSP) and the number of electric vehicle registrations, all have SHAP values below 3,500, with a relatively minor impact. In particular, the registration rate for electric vehicles in Australia is currently less than 3%, and has not yet had a significant impact on the overall curve of electricity demand. This might be related to the fact that the scale of electric vehicle ownership is still relatively small or that the charging time has not yet been highly concentrated.

Overall, these explanations verify the significance of the third-stage model: even after Prophet and SARIMA removed long-term trends and seasonal fluctuations, there are still significant short-term perturbation factors such as climate and energy prices affecting the model results. And SHAP provides a transparent and quantitative explanation for the sources of these disturbances.

5.4.2 Comparison Between Each Stage of Hybrid Models

Table 5.9 and Figure 5.12 comprehensively demonstrate the fitting effect of the hybrid model at each stage and the trend of its error changes. As can be seen from the table, as the modeling Stage progresses, the RMSE on the test set gradually decreases from 468,883.7 MWh in Stage 1 to 143,741.7 MWh in Stage 3, and the MAPE also significantly drops from 7.2% to 2.32%. The continuous optimization of the three-stage structure in terms of accuracy was verified.

Table 5.9: Comparison Between Each Stage of Hybrid Models

Stage	Test RMSE (MWh)	MAPE (%)
Stage1	468883.7	7.2
Stage2	252619.9	3.97
Stage3	143741.7	2.32

In Figure 5.12, the purple solid line represents the fitting situation between the final predicted value and the true value, which is significantly better than the models that only contain the trend (Stage 1: green dotdash line) or the trend + seasonal components (blue dotted line). It can be intuitively seen through the comparison of charts that the introduction of residual modeling (XGBoost) effectively captures nonlinear structures, making the model prediction closer to the real changes in electricity demand.

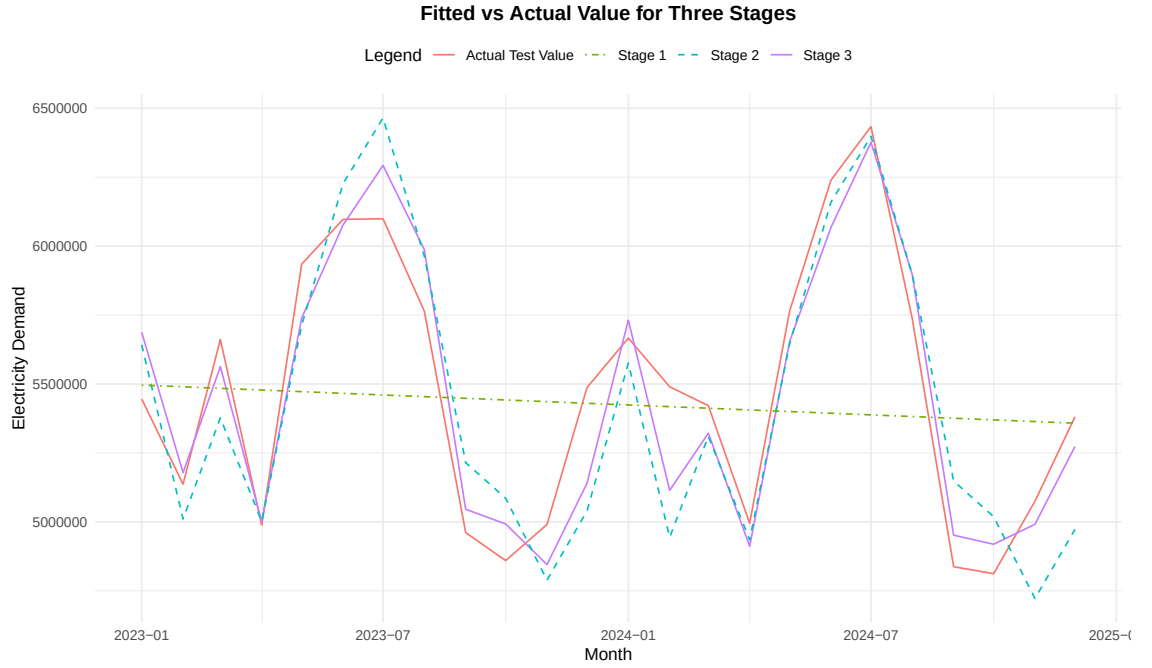


Figure 5.12: Comparison Between Each Stage of Hybrid Models

5.4.3 Comparison Between Individual and Hybrid Models

To evaluate the effectiveness of the proposed hybrid forecasting framework, this study compares its performance with several Individual models from three methodological categories:

- **Traditional time series methods:** Prophet and SARIMA
- **Machine learning methods:** XGBoost
- **Neural networks:** MLP and LSTM

Table 5.10: Comparison Between Individual and Hybrid Models

Types	Model	Test RMSE (MWh)	MAPE (%)
Traditional TS	Prophet	851,361.7	13.50
Traditional TS	SARIMA	207,701.2	3.28
Machine Learning	XGBoost	237,854.0	3.53
Neural Networks	MLP	658,202.2	10.00
Neural Networks	LSTM	492,917.7	7.76
Hybrid Model	Prophet + SARIMA + XGBoost	143741.7	2.32

As shown in Table 5.10, the **hybrid model (Prophet + SARIMA + XGBoost)** achieves the **lowest RMSE of 143,741.7** and **MAPE of 2.32%**, significantly outperforming all other individual models.

Among the individual models, SARIMA shows the best performance (RMSE = 207,701.2, MAPE = 3.28%), indicating its strength in capturing seasonal components. XGBoost also performs competitively with a MAPE of 3.53%, benefiting from structured feature learning. In contrast, Prophet yields the weakest performance (RMSE = 851,361.7, MAPE = 13.50%), likely due to its inability to model complex nonlinear relationships or residual patterns. MLP and LSTM, while better than Prophet, still fall behind SARIMA and the hybrid model.

These results demonstrate that integrating the trend (via Prophet), seasonal cycles (via SARIMA), and complex nonlinear residuals (via XGBoost) in a task-level hybrid architecture yields superior accuracy and robustness compared to any individual model.

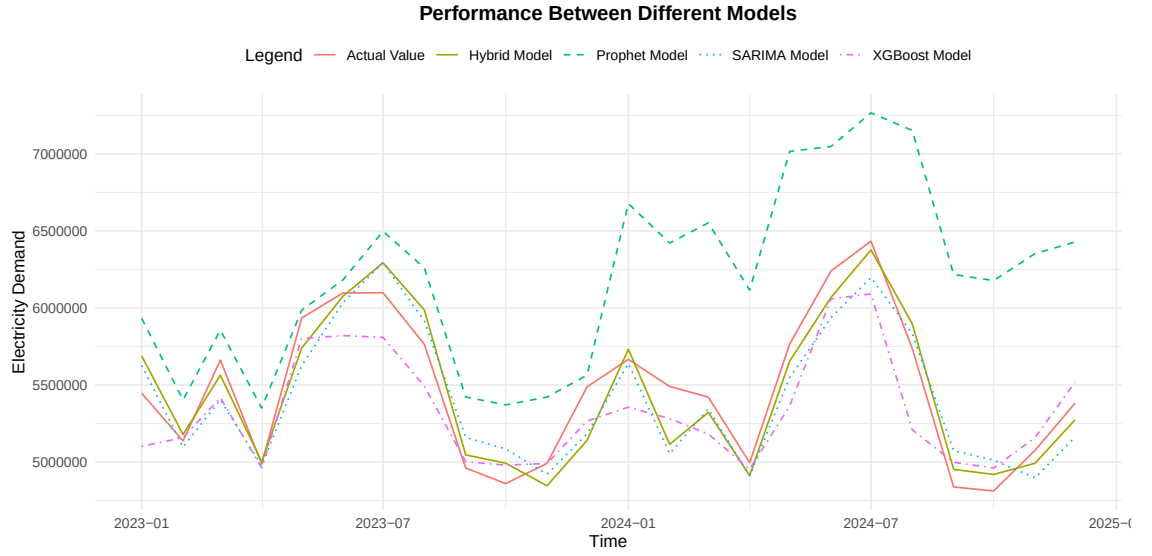


Figure 5.13: Comparison of Forecasting Curves by Various Models

Figure 5.13 illustrates the performance comparison among the Hybrid Model and Individual Models with the actual electricity demand in New South Wales from 2023 to 2025.

As shown in the plot, the Hybrid Model (orange line) closely follows the actual values (red line), especially during demand peaks and troughs. Prophet tends to overestimate during high-demand periods, while XGBoost and SARIMA exhibit slight underestimation in certain months.

Overall, the Hybrid Model demonstrates superior fitting performance compared to individual models, indicating its ability to integrate trend, seasonality, and residual dynamics more effectively.

5.5 Further Analysis

5.5.1 Performance of Different Season

To test the prediction stability of the model in different seasons, we divided the test set into four subsets: Summer, Autumn, Winter and Spring, and calculated the RMSE and MAPE of each group respectively. As shown in the figure 5.14 and figure 5.15, the model's performance in autumn and spring is relatively stable, with MAPE of 1.76% and 2.22% respectively. However, the errors are relatively high in summer and winter, with MAPE reaching 2.63% and 2.46% respectively. This might be related to the increased volatility of electricity demand under extreme weather conditions, especially the intensified air conditioning load in summer, making the prediction more challenging. Overall, the model remained within a reasonable error range in all four seasons and demonstrated good robustness.

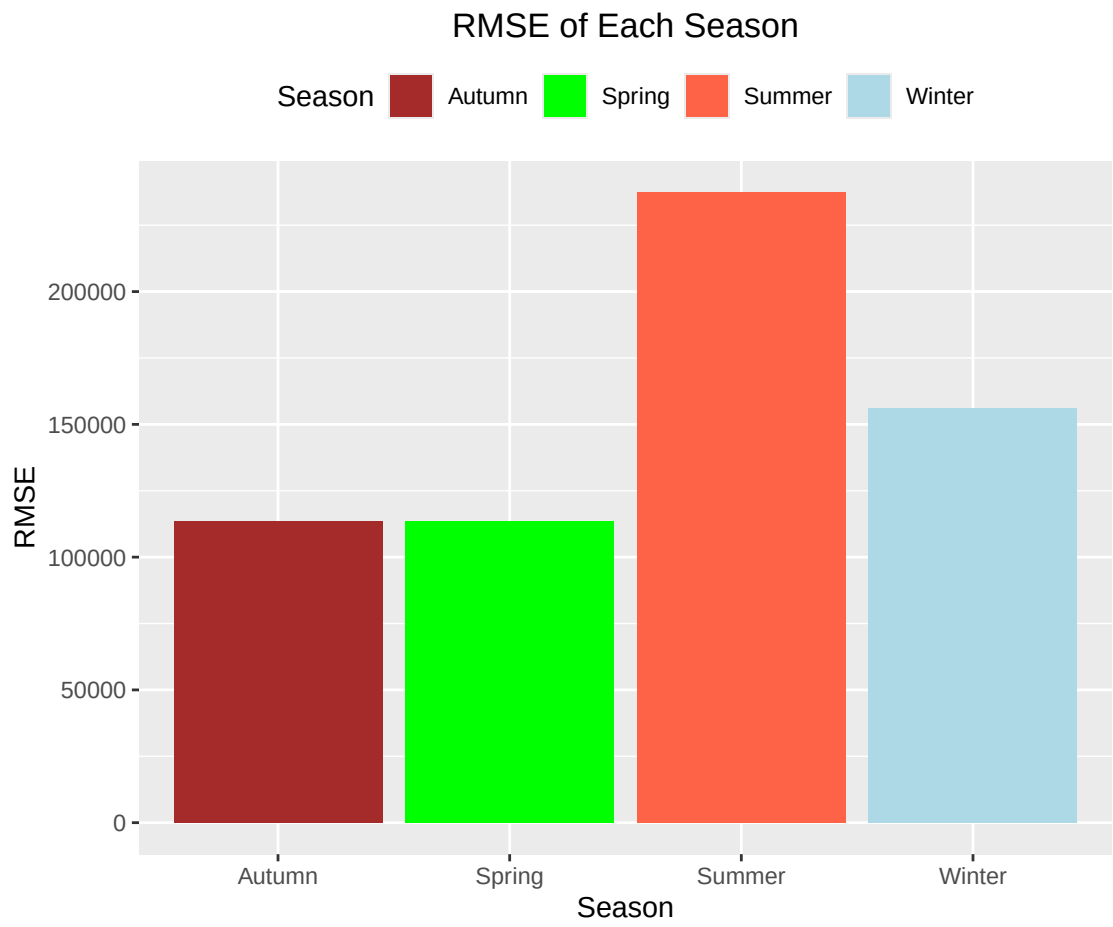


Figure 5.14: RMSE of Each Season

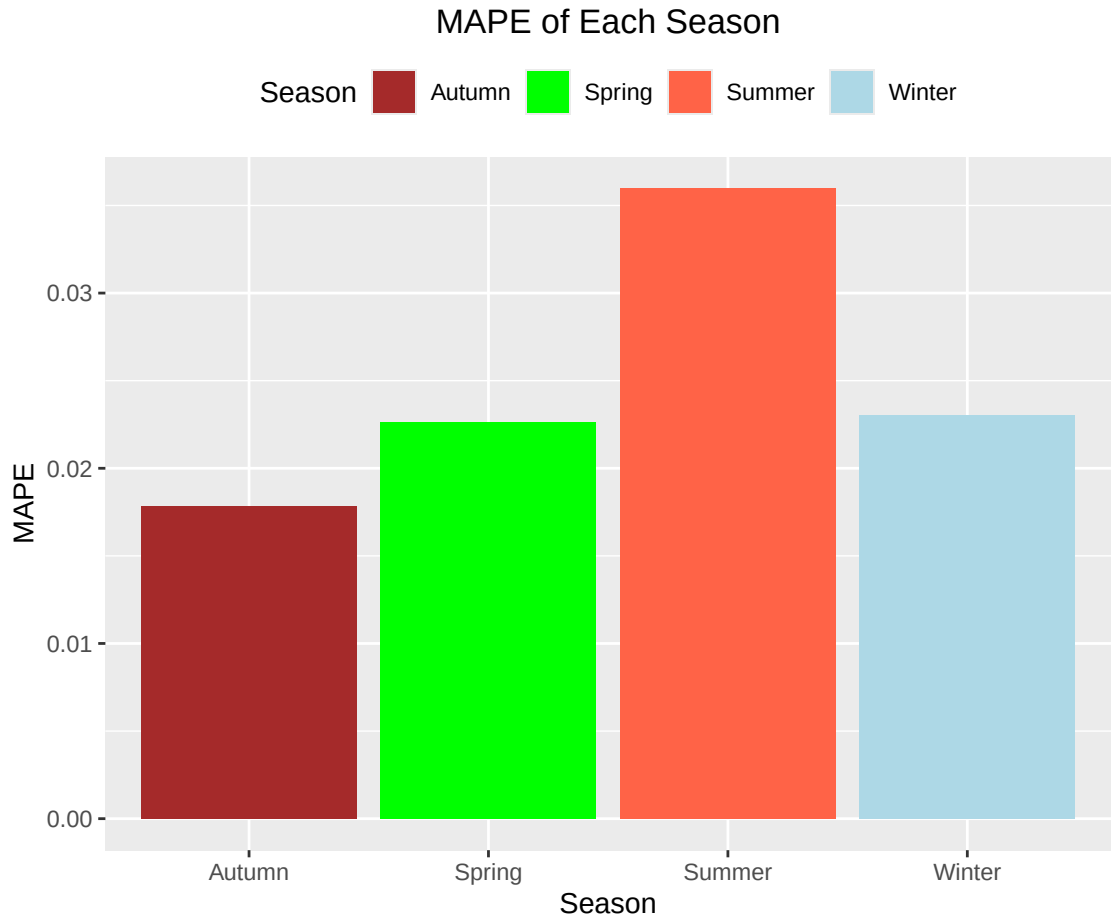


Figure 5.15: MAPE of Each Season

5.5.2 Monthly APE Trend Analysis

This section examines the distribution of the model's prediction error in different months by plotting the monthly APE (Absolute Percentage Error). The results show that the error between December and March of the following year is significantly high, especially in December 2023 and January 2024, with APE exceeding 6%. This period coincides with the Christmas and summer holidays in Australia and there are extreme fluctuations in electricity demand, which may make it difficult for the model to predict accurately. In addition, the errors in the remaining months were relatively stable, mostly controlled between 2% and 3%, indicating that the model has a good generalizability in most time periods. This analysis reveals the robustness issues of the model under extreme climate conditions, providing a direction for subsequent model optimization. For instance, feature variables such as holidays or temperature anomalies can be considered for modeling reinforcement.

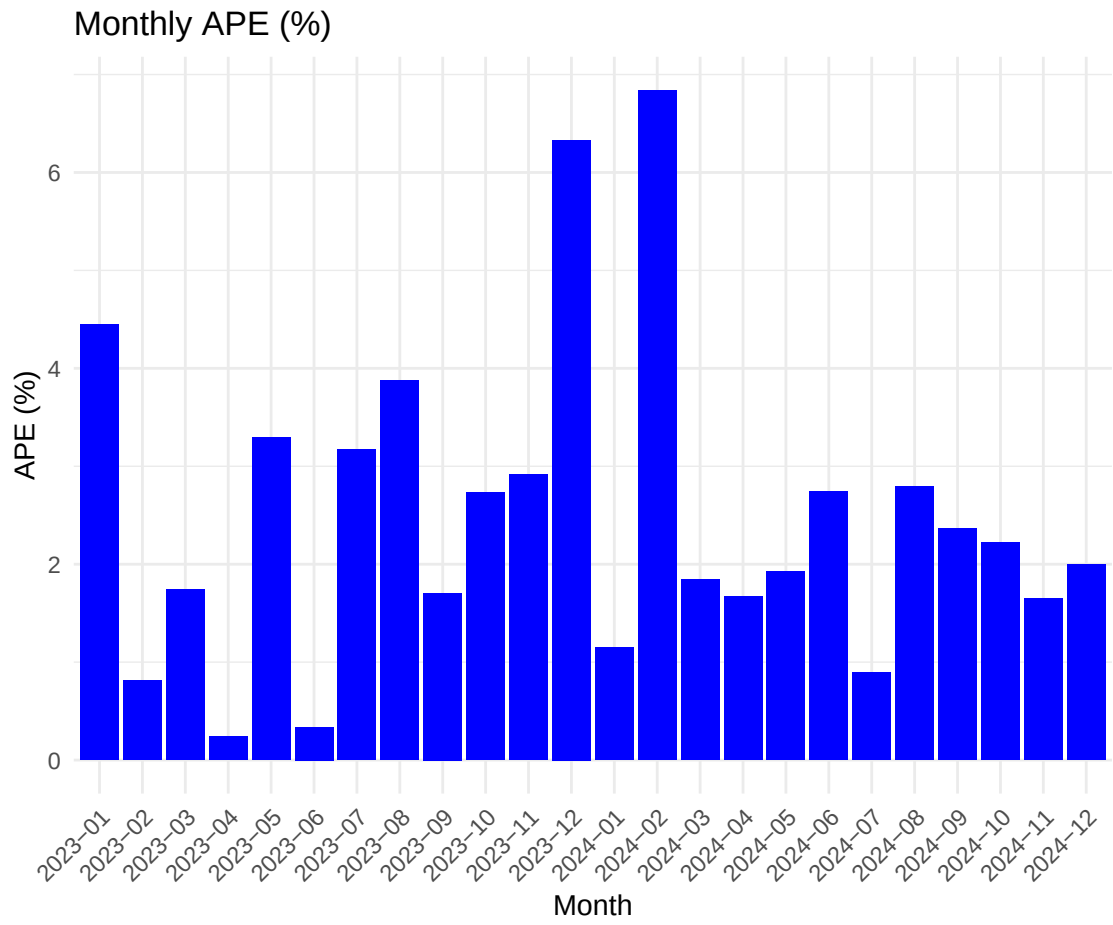


Figure 5.16: Monthly APE

CHAPTER 6

Conclusion and Future Work

6.1 Conclusion

This study focuses on the electricity load forecasting in New South Wales, Australia, with the goal of enhancing the prediction accuracy through optimizing the modeling methods, which can provide a scientific basis for electricity grid dispatching, energy supply and demand management, and policy making. The electricity demand is influenced by multiple factors, including economic gross product, periodic seasonal climate changes, and some unpredictable short-term events such as extreme weather. In this context, traditional single prediction models often struggle to simultaneously capture the complex nonlinear interactions among these factors. Therefore, constructing a hybrid modeling framework that can apply the most suitable model to different components has become an important direction for improving prediction accuracy.

Based on this approach, our study implemented a three-stage hybrid framework that decomposes the electric demand series into trend, seasonal and residual components. Each component was modeled using the most appropriate technique for its characteristics: Prophet for capturing long-term trends, SARIMA for modeling seasonal patterns, and XGBoost for predicting nonlinear residual fluctuations. To further improve interpretability, we applied SHAP analysis to explore the influence of input variables on the residual predictions.

A key innovation of our research is the use of a decomposed hybrid framework that assigns appropriate models to trend, seasonal, and residual components. The approach improves forecasting accuracy and overcomes the limitations of single model strategies.

Experimental results confirm the practical effectiveness of the proposed three-stage hybrid framework. It achieved a MAPE of 2.32% in forecasting electricity demand in New South Wales. This performance clearly outperforms single models, including Prophet (13.50%), SARIMA (3.28%), and XGBoost (3.53%). Compared with existing studies on electric load forecasting, such as the macro-ARIMA-LSTM model by González Grandón et al. (2023), which achieved a MAPE of 3.17%, and Eshragh et al. (2020) employed a structured time series regression model to forecast weekly peak electricity demand in NSW, achieving a MAPE of approximately 3.0%. Our framework realised a lower error rate, demonstrating its practical advantage in long-term forecasting.

6.2 Limitations and Future Work

Although the proposed framework performed well overall, it still faces challenges related to component separation, incomplete feature representation, and model or

parameters tuning. Specifically, the decomposition step may not achieve a complete separation of trend and seasonal signals, which could result in residual components still containing trends or seasonality component. Additionally, the model does not consider certain external factors, such as electricity pricing or policy interventions, which played a significant role but were excluded due to limited data availability. Lastly, although cross-validation was employed for parameter tuning, further improvements in forecasting accuracy might be achieved through more advanced optimization methods.

To enhance the current approach, several directions for improvement can be considered. Applying more sophisticated methods like VMD may improve the clarity of decomposition and reduce interference between components. This can be validated by checking the ACF of the residuals to see whether any structure remains. In addition, Adding external variables like policy changes, energy prices, or industrial activity could offer additional explanatory power and help the model better capture the key factors of electricity demand. On the modeling side, there's still room to improve how residuals are modeled, whether by adjusting the tuning process or exploring different algorithms.

Furthermore, seasonal error analysis indicates that the model performs relatively poorly in extreme seasons such as summer and winter. Although the prediction error is still within an acceptable range, the deviation indicates that the model fails to fully capture the changing patterns of electricity demand when facing highly uncertain scenarios such as large temperature fluctuations. In future research, it is possible to consider introducing seasonally specific modeling mechanisms or external variables related to extreme weather to optimize the peak seasonal performance of the model.

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